SHEAF BANDWIDTH:
A CATEGORICAL BRIDGE BETWEEN TOPOLOGICAL SIGNAL PROCESSING
AND FINITE ELEMENT EXTERIOR CALCULUS

[AUTHOR PLACEHOLDER]

ABSTRACT. A cellular sheaf of Hilbert spaces equips a regular cell complex with stalks and restriction maps, and Robinson [Rob14b, §6] introduced an analytic invariant of such sheaves—the *bandwidth*—while leaving open whether this invariant admits a definition intrinsic to the sheaf rather than to a chosen ambient representation. We propose $\mathrm{sheafBW}(\mathscr{A}) := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$, prove unitary invariance (Theorem 5.5), and establish a strict equality $\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) = \sigma_{\min}(A)/\sigma_{\max}(A)$ (Theorem 5.19) on an abstract sampling cell complex X^A with a canonical double 0-cell construction. The framework is instantiated in three previously disjoint domains—topological signal processing on simplicial complexes, finite element electromagnetics, and Brillouin zone tetrahedron quadrature in condensed matter—unified by a common categorical template, with a triangle equivalence (Theorem 3.1) connecting finite element exterior calculus singular value bounds, topological signal processing Riesz frame conditions, and Robinson sheaf cohomology; numerical verification across three toy operators matching the common finite-dimensional template confirms the strict equality to within the expected floating-point error budget (reldiff $\leq 1.64 \times 10^{-14}$, driven by the squaring of the condition number under the A^*A form). An empirical proposition (Theorem 6.3) relates the sampling and integration maps via a Pareto trade-off, supported by two empirical channels (a project-internal numerical study and five decades of Brillouin zone practice) and a Fourier-spectrum heuristic. Two independent open problems remain: a non-trivial dagger structure on the full category (Theorem 5.25) and a constructive Pareto proof of the empirical proposition (Theorem 6.4).

1. Introduction

1.1. **Robinson's open problem on sheaf bandwidth.** Cellular sheaves of Hilbert spaces, as developed by [Rob14b] and refined in the spectral theory of [HG19b], equip a regular cell complex X with stalks $\mathscr{A}(\sigma)$ on each cell and bounded restriction maps along incidences. Robinson's monograph closes [Rob14b, §6] with an explicit open question: does sheaf bandwidth admit a definition that is intrinsic to the sheaf, not merely to a chosen ambient representation? In our reading, the question splits into three sub-questions:

(Q1) **Categorical intrinsicness.** Is there a real-valued invariant $\mathrm{sheafBW}(\mathscr{A})$, defined from the sheaf data alone (stalks, restriction maps, sheaf Laplacian), that is preserved under unitary equivalence in the ambient cellular sheaf category?

(Q2) **Spectral handle.** Does this invariant admit a closed form on the spectrum of the sheaf Laplacian $L^0 = (\delta^0)^*\delta^0$, comparable in tractability to the singular value ratio $\sigma_{\min}/\sigma_{\max}$ of an ambient matrix?

(Q3) **Concrete reduction.** On a canonical class of sheaves arising from a sampling problem, does the intrinsic invariant reduce *strictly* (not just up to constants) to a domain-specific singular value ratio?

Date: May 17, 2026.

2020 *Mathematics Subject Classification.* 18F20, 65N30, 94A20, 55U10.

Key words and phrases. Sheaf bandwidth, cellular sheaf, finite element exterior calculus, Hodge decomposition, sampling theorem, Brillouin zone tetrahedron, k-space sampling, MRI.

This paper answers all three affirmatively, in the finite-dimensional, spectral-gap, unitary-invariant version of the question as split here (Theorems 5.1, 5.5 and 5.19), and exhibits the answer in three independent application domains (section 6). The technical formulation in section 3.6 narrows these three sub-questions to a working two-clause specification, in which (Q2) is absorbed into the spectral-gap form of Theorem 5.1 itself. The infinite-dimensional extension and any categorical generalization beyond the present cellular-sheaf-of-finite-dimensional-Hilbert-spaces setting are explicitly out of scope; see section 3.7 for the four boundary conditions (L1)–(L4).

1.2. **The black-box treatment of k-space sampling in MRI.** Magnetic resonance imaging reconstruction by 2026 rests on four schools that treat the choice of sampling geometry as a black box upstream of reconstruction:

- *Compressed sensing* [LDP07] treats sampling as a fixed acquisition pattern and optimizes only the reconstruction prior;
- *Topological signal processing* [BS20] treats sampling on simplicial complexes via Riesz frame inequalities, again with the geometry given;
- *Sparkling-style point process design* [LWC+19] treats sampling design as a point-process optimization detached from the reconstruction operator;
- *Sheaf signal processing* [Rob14b, HG19b, HG19a] studies sheaf bandwidth as an analytic invariant without exposing a computable reduction to acquisition-side singular values.

Each school is internally precise. None exposes a single scalar that is at once (i) intrinsic to the abstract sampling structure and (ii) directly computable from the acquisition matrix. We will refer to this gap as the *sampling black box* and resolve it via the triangle equivalence of Theorem 3.1.

1.3. **The FEEC and cellular-sheaf disconnect.** Finite element exterior calculus (FEEC), as systematized by [AFW10, FW14] on the algebraic side and by [BH06, Chr10] on the construction side, provides closed Hilbert complex theory, Hodge decomposition with stalk-wise harmonic representatives, and bounded cochain projections of arbitrary commuting type. Cellular sheaf signal processing, beginning with [Rob14b], develops cohomology and bandwidth on the same combinatorial substrate. Despite the shared substrate of cell complexes equipped with cochain spaces, the two literatures have not been merged in print: [CR25] surveys 30 years of FEEC without referencing sheaf signal processing, and [CKL25] surveys recent topological signal processing without invoking AFW Hilbert complexes. We attribute this to a missing concrete bridge—a single object on which the two machineries meet and produce the same scalar—which we supply via the abstract sampling cell complex X^A of Theorem 5.12.

1.4. **Contributions.** The contributions of this paper are the following.

(C1) **Triangle Equivalence Theorem (Theorem 3.1).** Three formulations of well-conditioning for k-space sampling—FEEC singular value bounds, topological signal processing Riesz frame conditions, and Robinson sheaf cohomology with sheaf bandwidth—are formally equivalent *under the common finite-dimensional sampling-operator normal form* of section 3.1; the equivalence is not claimed to hold in the original infinite-dimensional or qualitative settings of the three source literatures. The proof of the equivalence is decomposed into a linear-algebraic part $(a) \Leftrightarrow (b)$ (section 3.3) and a sheaf-cohomological part $(a) \Leftrightarrow (c)$ (section 3.4).

(C2) **Sheaf bandwidth, categorically intrinsic (Theorems 5.1 and 5.5).** The proposed invariant $\mathrm{sheafBW}(\mathscr{A}) := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$ uses only the sheaf Laplacian spectral gap and is unitarily invariant in the cellular sheaf category. This addresses sub-questions (Q1) and (Q2) of section 1.1.

(C3) **Strict equality on the abstract sampling cell complex (Theorem 5.19).** On the canonical sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ over the cell complex X^A (Theorems 5.12 and 5.14), constructed with a double 0-cell $\{\sigma_B, \sigma_\infty\}$ to satisfy the regularity conditions of [HG19b, Def. 1, Cond. 4], the intrinsic invariant reduces strictly to $\sigma_{\min}(A)/\sigma_{\max}(A)$ on the well-conditioned locus $\sigma_{\min}(A) > 0$ (equivalently $H^0(\mathscr{A}^{\text{samp}}) = 0$); the rank-deficient boundary is recorded in Theorem 5.21. This addresses sub-question (Q3) of section 1.1.

(C4) **Three-instance unification by a common categorical template (section 6).** Three previously disjoint domains fit the same template $V_? \subset \mathbb{C}^N + A : V_? \to \mathbb{C}^M$: topological signal processing ($V_? = V_B^{\text{TopSP}}$, the bandlimited subspace), finite element electromagnetics ($V_? = V_S^{\text{FEM}}$, a finite-dimensional restriction of an AFW conforming subspace), and Brillouin zone tetrahedron quadrature ($V_? = V_{\text{occ}}^{\text{BZ}}$, the Whitney 0-form span over occupied k-points; see section 6.4 for scope). Theorem 5.19 applies verbatim in each.

(C5) **Empirical proposition on sampling–integration trade-off (Theorem 6.3).** The sampling map S (Theorem 6.1) and the integration map I (Theorem 6.2) trace a Pareto frontier under geometry optimization: no admissible V simultaneously maximizes S and minimizes I. The supporting evidence comprises two empirical channels and a heuristic skeleton: a project-internal numerical study on 35 MRI sampling configurations with a multi-N multi-K robustness stratification (section 7); five decades of Brillouin zone practice in condensed matter [BJA94, LT72, JA71, MP76], on the integration side; and the Euclidean Fourier-spectrum framework of [PSC+15], contributing directional content only rather than direct evidence on cellular-sheaf integration.

(C6) **Numerical verification of the strict equality (section 7).** The strict equality of Theorem 5.19 is verified across three toy operators matching the common finite-dimensional template, with maximum relative deviation 1.64×10^{-14} (Instance 2; Instances 1 and 3 are within 8×10^{-16}). The deviation reflects the expected roundoff amplification of comparing $\sqrt{\lambda_{\min}(A^*A)/\lambda_{\max}(A^*A)}$ with $\sigma_{\min}(A)/\sigma_{\max}(A)$ on conditioned matrices, since $\kappa(A^*A) = \kappa(A)^2$. The verification is honest in scope: it confirms the linear-algebraic identity and the code path, not the underlying physical evidence per instance, which is recorded separately in table 6.

(C7) **Two independent open problems (Theorems 5.25 and 6.4).** The categorical bridge constructed here leaves two precisely formulated open questions: a non-trivial dagger structure on the full category $\mathsf{Cell\text{-}Sh}^{\text{fin}}_{\mathsf{Hilb}}(X)$ extending the unitary-subcategory groupoid (Theorem 5.23); and a constructive Pareto-optimality proof upgrading Theorem 6.3 to a theorem via compactness, convex relaxation, and spectral duality.

1.5. **Paper outline.** Section 2 reviews the four schools whose tools we combine: finite element exterior calculus, cellular sheaf signal processing, topological signal processing, and dagger Galerkin theory. Section 3 states and proves the triangle equivalence theorem and produces a candidate sheaf bandwidth that is not yet categorically intrinsic. Section 4 reviews the FEEC machinery (closed Hilbert complex, Hodge decomposition, bounded cochain projection) needed to lift bandwidth to a categorically intrinsic invariant, and presents a four-school comparison table. Section 5 constructs the categorically intrinsic candidate sheafBW, proves unitary invariance, builds the abstract sampling cell complex X^A with its double 0-cell, and establishes the strict equality Theorem 5.19. Section 6 instantiates the framework in topological signal processing, finite element electromagnetics, and Brillouin zone quadrature, states the empirical proposition on the S–I Pareto trade-off, and records the constructive Pareto proof as an independent open problem. Section 7 verifies the strict equality numerically and states the scope of the verification. Section 8 restates the seven contributions, discusses the two open problems, and outlines the engineering verification on the MRI side that lies outside this paper.

The three motivating threads—Robinson's open question, the MRI black-box treatment of sampling, and the FEEC and cellular-sheaf disconnect—are addressed jointly by a single object on which all three meet. Section 2 reviews the four schools whose tools we use, and section 3 begins the construction with a triangle equivalence theorem.

Formalization in Lean 4. The main finite-dimensional operator/sheaf-core theorems, propositions, and corollaries of section 3 and section 5 are formalized in Lean 4 (mathlib `v4.30.0-rc2`) and machine-verified by the Lean kernel: `lake build` passes the entire library with *0 sorry, 0 admitted, 0 warnings* (8326 build jobs). Each formalized result carries a footnote identifying the corresponding Lean file and theorem name(s). The category-theoretic subsection section 5.7 (i.e., Theorems 5.22 and 5.23), the physical instances of section 6 (i.e., Theorems 6.1 and 6.2 and Theorem 6.3), and the geometric "scalar multiple of isometry" clause of Theorem 5.19 are *not* formalized; each such statement carries an explicit "Not formalized" footnote at its declaration. The complete paper–Lean correspondence table is in `lean/CROSSREF.md`.

2. Related Work

The technical work in this paper combines tools from four schools that have developed independently. Each school is internally precise; none has been combined with the others on a single concrete object producing a single scalar. We review the relevant results here so that later sections can cite them by tag.

2.1. **Finite element exterior calculus.** Finite element exterior calculus rests on three lines of construction. The reduction–reconstruction interface of [BH06] (with antecedents in [Whi57, Bos88]) defines a map $R : \Lambda^k(\Omega) \to C^k$ that integrates a smooth k-form over chains, and an inverse $I_W : C^k \to \Lambda^k$ that lifts chains to forms via Whitney shape functions; these two maps satisfy a commuting Stokes identity (Theorem 4.2). The finite-element-system framework of [Chr10] packages these data abstractly: an FES is a contravariant functor on a category of cellular complexes, and Whitney forms are recovered as its canonical instance. The Hilbert complex framework of [AFW10, FW14] (refining [BL92, AFW06]) supplies the analytic core that we need: closed cochain complexes of Hilbert spaces, Hodge decomposition with stalk-wise harmonic representatives, and bounded cochain projections π_h commuting with the differential up to a fixed constant. The recent survey of [CR25] collates 30 years of FEEC development and is one of our two anchors for placing FEEC tools relative to the topological-signal-processing literature.

2.2. **Cellular sheaves and sheaf signal processing.** Cellular sheaves of vector spaces, with cohomology computed by a chain complex on a regular cell complex, are surveyed in [Rob14b, §6]. Robinson defines sheaf bandwidth as a real-valued analytic invariant and closes the chapter with the open question on its categorically intrinsic formulation that motivates the present paper. An earlier sheaf-theoretic perspective on sampling, complementary in topic to the bandwidth question studied here, appears in [Rob14a]; the abstract sampling cell complex X^A and the strict-equality construction of Theorem 5.19 are independent of that work, supplying a specific finite-dimensional sampling sheaf on which Robinson's open question admits a concrete spectral resolution. The spectral theory of cellular sheaf Laplacians $L^k = (\delta^k)^*\delta^k + \delta^{k-1}(\delta^{k-1})^*$ is developed in [HG19b]; we use Definition 1 of that paper, especially Cond. 4 (boundaries of 1-cells consist of two distinct 0-cells), to certify regularity of the abstract sampling cell complex of Theorem 5.12. The conference version [HG19a] introduces the spectral gap quantities γ_+, Γ_+ on which our intrinsic invariant sheafBW $:= \sqrt{\gamma_+/\Gamma_+}$ is built.

2.3. **Topological signal processing.** Topological signal processing on simplicial complexes is the program initiated by [BS20]: signals live on k-cochains, the simplicial Laplacian replaces the graph Laplacian, and bandlimited subspaces are defined as eigenspaces of the simplicial Laplacian below a fixed threshold B. The school treats sampling and reconstruction via Riesz frame inequalities in the bandlimited subspace, an angle that we recover as condition (b) of Theorem 3.1. The recent survey of [CKL25] is our second anchor for placing topological signal processing relative to FEEC; that survey does not invoke AFW Hilbert complexes.

2.4. **Dagger Galerkin theory.** A categorical formulation of Galerkin theory in which monomorphisms carry a † (right-adjoint) structure is given by [LS20]; the abstract dagger setting is [HV19]. We use this thread sparingly: the dagger tools clarify which subcategories of $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)$ carry a unitary structure, and they make Theorem 5.25 a precise question about the existence of a non-trivial † on the full category. The unitary subcategory $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)^{\mathrm{u}}$ of Theorem 5.22 is a groupoid (Theorem 5.23); the question is whether this restricted structure extends.

2.5. **What has not been combined.** Each of the four schools above provides precise tools, and each admits the underlying combinatorial object—a regular cell complex with cochain spaces and a coboundary operator—without difficulty. Yet the four have not been combined on a single concrete object producing a single scalar that is at once intrinsic to the abstract structure and computable from acquisition data. The references cited in this section are mutually disjoint: [AFW10] does not invoke cellular sheaves; [HG19b] does not invoke FEEC; [BS20] does not invoke either; [LS20] does not invoke any signal processing applications. This paper supplies the missing object (the abstract sampling cell complex X^A of Theorem 5.12) and the missing scalar (the strict equality of Theorem 5.19). Table 1 in section 4 records the four-school comparison once the needed FEEC machinery has been reviewed.

Each school provides precise tools but operates in isolation. We now exhibit the first formal bridge between three of them via a triangle equivalence theorem (section 3), which pinpoints exactly where Robinson's open problem lives.

3. Triangle Equivalence: FEEC Geometry, Hodge Spectrum, and Robinson Sheaf Cohomology

We now construct the bridge promised in section 1: three formulations of well-conditioning for k-space sampling, drawn from three schools that have not previously been combined—finite element exterior calculus (FEEC) singular value bounds, topological signal processing (TopSP) Riesz frame conditions, and Robinson sheaf cohomology with sheaf bandwidth—are formally equivalent under the common finite-dimensional sampling-operator normal form introduced in section 3.1. The equivalence is asserted in this finite-dimensional normal form; the original infinite-dimensional or qualitative formulations of the three source literatures are not themselves claimed to be equivalent outside this normalization (see section 3.7, (L1)). The equivalence is stated as Theorem 3.1 and proved in sections 3.3 and 3.4. The proof of $(a) \Leftrightarrow (c)$ produces a candidate definition of sheaf bandwidth that is not categorically intrinsic; the categorically intrinsic version is constructed in section 5. We close the section by formalizing Robinson's open problem from [Rob14b, §6] as a precisely narrowed two-clause question (section 3.6).

3.1. **Setup and notation.** The well-conditioning of k-space sampling admits three equivalent readings: an SVD-style ratio $\sigma_{\min}(A)/\sigma_{\max}(A)$ of the sampling operator; a Riesz frame inequality on a bandlimited subspace; and a sheaf-cohomological condition on an abstract sampling cell complex. Theorem 3.1 makes this precise: the three readings agree on the same threshold $\tau \in (0, 1]$. The setup below introduces the common notation in which all three formulations live—the bandlimited subspace V_B, the sampling operator A, and the condition number $\kappa(A)$—before the triangle statement in section 3.2.

Let $\Omega_k := B(0, k_{\max}) \subset \mathbb{R}^d$ be a closed ball in k-space, and let $V = \{k_1, \ldots, k_N\} \subset \Omega_k$ be a finite set of N candidate sample positions. We do not impose geometric structure on V at this stage; the dependence of the constructions on V is recorded explicitly via $V_B(V)$ below and discussed in section 6.

Let $X(V)$ denote the Delaunay simplicial complex on V (cells: vertices V, edges E, faces F, tetrahedra T), oriented in the sense of [Hat02, §2.1]. Edge weights are $w_{ij} := |k_i - k_j|^{-2}$ (with the convention $w_{ij} = 0$ if $\{i, j\} \notin E$), and $W := (w_{ij})_{1 \le i,j \le N}$. Set $D := \mathrm{diag}(d_1, \ldots, d_N)$ with

$d_i := \sum_j w_{ij}$, and define the graph Laplacian $L_0 := D - W \in \mathbb{R}^{N\times N}$. The eigenpairs of L_0 are denoted $(\mu_l, v_l)_{l=0,\dots,N-1}$ with $0 = \mu_0 \le \mu_1 \le \cdots \le \mu_{N-1}$ and $\{v_l\}_{l=0}^{N-1}$ an orthonormal basis of $\mathbb{C}^N$.

For a fixed bandwidth threshold $B \in (0, \mu_{N-1})$, define the B-bandlimited subspace

$$V_B(V) := \operatorname{span}\{\, v_l : \mu_l \le B \,\} \;\subset\; \mathbb{C}^N, \qquad K := \dim V_B(V). \tag{1}$$

We write V_B for $V_B(V)$ when V is fixed and emphasize the V-dependence only when comparing across instances (section 6).

Fix a sampling subset $V_s \subset \{1,\dots,N\}$ with $M := |V_s| \ge K$. Let $\Sigma_{V_s} : \mathbb{C}^N \to \mathbb{C}^M$ be the index restriction operator $(\Sigma_{V_s} c)_n := c_{i_n}$ for $V_s = \{i_1,\dots,i_M\}$, and let $\iota_B : V_B \hookrightarrow \mathbb{C}^N$ be the canonical isometric embedding. The *sampling operator*

$$A \;:=\; \Sigma_{V_s} \circ \iota_B \;:\; V_B \longrightarrow \mathbb{C}^M \tag{2}$$

is the central object of this paper. In the basis $\{v_l\}_{l=0}^{K-1}$ of V_B and the canonical basis of $\mathbb{C}^M$, A has the matrix representation

$$A \;=\; \big[\, v_l(i_n) \,\big]_{n=1,\dots,M;\, l=0,\dots,K-1} \;\in\; \mathbb{C}^{M\times K}.$$

The singular values of A are denoted

$$\sigma_{\max}(A) \;:=\; \sigma_0(A) \;\ge\; \sigma_1(A) \;\ge\; \cdots \;\ge\; \sigma_{K-1}(A) \;=:\; \sigma_{\min}(A) \;\ge\; 0.$$

The condition number is $\kappa(A) := \sigma_{\max}(A)/\sigma_{\min}(A) \in [1,\infty]$, with $\kappa(A) = \infty$ when $\sigma_{\min}(A) = 0$ (rank deficiency). Throughout, $\tau \in (0,1]$ denotes a fixed well-conditioning threshold; A is *τ-well-conditioned* when $\kappa(A) \le 1/\tau$, equivalently $\sigma_{\min}(A) \ge \tau\, \sigma_{\max}(A)$.

3.2. **Triangle equivalence theorem.**

Theorem 3.1 (Triangle Equivalence)**.** [1] *Fix V, V_s, B (and hence $V_B = V_B(V)$ and A as in section 3.1), and fix $\tau \in (0,1]$. The following three statements are equivalent.*

(a) (FEEC singular value form.) $\sigma_{\min}(A) \ge \tau\, \sigma_{\max}(A)$.

(b) (Hodge-spectrum Riesz form.) *There exist constants $0 < \alpha \le \beta$ with $\beta/\alpha \le 1/\tau$ such that for every $c \in V_B \setminus \{0\}$,*

$$\alpha\, \|c\|_{\mathbb{C}^N} \;\le\; \|\Sigma_{V_s} c\|_{\mathbb{C}^M} \;\le\; \beta\, \|c\|_{\mathbb{C}^N}. \tag{3}$$

That is, V_B forms a Riesz sequence in $\ell^2(V_s)$ with frame bound ratio $\beta/\alpha \le 1/\tau$.

(c) (Robinson sheaf form.) *On the* abstract sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ *over the abstract sampling cell complex X^A, constructed in Theorems 5.12 and 5.14 of section 5,*

$$H^0\big(X^A; \mathscr{A}^{\mathrm{samp}}\big) \;=\; 0 \qquad \textit{and} \qquad \mathrm{sheafBW}^{\mathrm{cand}}(\mathscr{A}^{\mathrm{samp}}) \;\ge\; \tau, \tag{4}$$

where $\mathrm{sheafBW}^{\mathrm{cand}}$ *is the candidate sheaf bandwidth of Theorem 3.3 below.*

Remark 3.2 (Three readings of one inequality)**.** The three forms in Theorem 3.1 read the same scalar inequality in three vocabularies developed in disjoint literatures: the FEEC singular value form is the standard reading via singular value bounds [AFW10]; the Riesz form is the TopSP reading via ℓ^2 frame bounds [BS20]; the Robinson form translates the inequality into the language of cellular sheaves on $X(V)$ in the sense of [Rob14b], with the sheaf bandwidth furnishing the quantitative bound.

[1] Formalized in Lean 4 as `triangle_equivalence` in `lean/MRI/Cellular/TriangleEquivalence.lean`. The Lean statement formalizes the finite-dimensional operator/sheaf core of the theorem, with V_B represented by an abstract Hilbert space and A by an evaluation operator; the construction of $V_B(V)$ from the Delaunay graph is treated as setup data rather than separately formalized geometry. Full correspondence: `lean/CROSSREF.md`.

Theorem 3.1 is proved in two parts: $(a) \Leftrightarrow (b)$ in section 3.3, and $(a) \Leftrightarrow (c)$ in section 3.4. The latter relies on the abstract sampling sheaf $\mathscr{A}^{\text{samp}}$ over the cell complex X^A, whose explicit construction is given in section 5.5 of section 5; the candidate sheaf bandwidth $\text{sheafBW}^{\text{cand}}$ used in clause (c) is recorded in Theorem 3.3 below in terms of $\mathscr{A}^{\text{samp}}$.

3.3. **Proof of** $(a) \Leftrightarrow (b)$**.** The equivalence $(a) \Leftrightarrow (b)$ is a direct consequence of the variational characterization of singular values together with the isometry of ι_B.

Proof of $(a) \Leftrightarrow (b)$*.* Since $\iota_B : V_B \hookrightarrow \mathbb{C}^N$ is an isometric embedding, $\|\iota_B c\|_{\mathbb{C}^N} = \|c\|_{V_B}$ for every $c \in V_B$, and consequently

$$\|Ac\|_{\mathbb{C}^M} \;=\; \|\Sigma_{V_s}\iota_B c\|_{\mathbb{C}^M} \;=\; \|\Sigma_{V_s} c\|_{\mathbb{C}^M} \qquad \text{for every } c \in V_B, \tag{5}$$

where in the last equality V_B is identified with its image $\iota_B(V_B) \subset \mathbb{C}^N$. The singular values of A are then characterized variationally by

$$\sigma_{\max}(A) \;=\; \max_{\substack{c\in V_B\\ \|c\|=1}} \|\Sigma_{V_s} c\|_{\mathbb{C}^M}, \tag{6}$$

$$\sigma_{\min}(A) \;=\; \min_{\substack{c\in V_B\\ \|c\|=1}} \|\Sigma_{V_s} c\|_{\mathbb{C}^M}. \tag{7}$$

$(b) \Rightarrow (a)$. Given α, β with the bounds eq. (3), take infimum and supremum over $c \in V_B$ with $\|c\| = 1$. From eqs. (6) and (7) this yields $\sigma_{\max}(A) \le \beta$ and $\sigma_{\min}(A) \ge \alpha$. The hypothesis $\beta/\alpha \le 1/\tau$ then gives $\sigma_{\min}(A)/\sigma_{\max}(A) \ge \alpha/\beta \ge \tau$, which is (a).
$(a) \Rightarrow (b)$. Set $\alpha := \sigma_{\min}(A)$ and $\beta := \sigma_{\max}(A)$. Then $\beta/\alpha = \kappa(A) \le 1/\tau$ by hypothesis. Equations (6) and (7) together with eq. (5) yield, after homogeneity in c, the bound eq. (3) with these constants. □

3.4. **Proof of** $(a) \Leftrightarrow (c)$**.** The equivalence $(a) \Leftrightarrow (c)$ is realized through the abstract sampling sheaf $\mathscr{A}^{\text{samp}}$ on the cell complex X^A, whose explicit construction we forward-reference to Theorems 5.12 and 5.14 in section 5.5. Two facts about $\mathscr{A}^{\text{samp}}$ proved there suffice for the present argument:

(F1) *Cochain spaces.* $C^0(\mathscr{A}^{\text{samp}}) \cong V_B$ (canonical isometry, Theorem 5.16) and $C^1(\mathscr{A}^{\text{samp}}) = \mathbb{C}^M$ (Theorem 5.17).

(F2) *Coboundary equals sampling operator.* The 0-th coboundary $\delta^0 : C^0(\mathscr{A}^{\text{samp}}) \to C^1(\mathscr{A}^{\text{samp}})$ of $\mathscr{A}^{\text{samp}}$ coincides with A after the canonical identification of Theorems 5.16 and 5.17, i.e. $\delta^0 = A$ as linear maps $V_B \to \mathbb{C}^M$ (Theorem 5.18).

Both facts depend only on the data of Theorems 5.12 and 5.14 and on the sampling operator A recorded in section 3.1; neither depends on Theorem 3.1, so the forward references are not circular.

Definition 3.3. The *candidate sheaf bandwidth* of $\mathscr{A}^{\text{samp}}$ is

$$\text{sheafBW}^{\text{cand}}(\mathscr{A}^{\text{samp}}) \;:=\; \frac{\sigma_{\min}(A)}{\sigma_{\max}(A)} \;\in\; [0,1], \tag{8}$$

with the convention $\text{sheafBW}^{\text{cand}}(\mathscr{A}^{\text{samp}}) := 0$ when $\sigma_{\min}(A) = 0$. This candidate references $\mathscr{A}^{\text{samp}}$ through the operator A derived from the chosen embedding $V_B \hookrightarrow \mathbb{C}^N$ and the basis $\{v_l\}_{l=0}^{K-1}$; it is therefore not categorically intrinsic to the cellular sheaf data of $\mathscr{A}^{\text{samp}}$. The categorically intrinsic version is the spectral-gap ratio $\text{sheafBW} := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$ constructed in section 5, which on $\mathscr{A}^{\text{samp}}$ satisfies a strict identity with $\sigma_{\min}(A)/\sigma_{\max}(A)$ via Theorem 5.19.

Proof of $(a) \Leftrightarrow (c)$*.* By (F1), the coboundary δ^0 of $\mathscr{A}^{\text{samp}}$ is well-typed as a linear map $V_B \to \mathbb{C}^M$, so the identification with A in (F2) is type-compatible. Combining (F1) and (F2) gives $\delta^0 = A$ as maps $V_B \to \mathbb{C}^M$, hence $H^0(\mathscr{A}^{\text{samp}}) = \ker \delta^0 = \ker A$, and so $H^0(\mathscr{A}^{\text{samp}}) = 0$ if and only if $\sigma_{\min}(A) > 0$. By Theorem 3.3, $\text{sheafBW}^{\text{cand}}(\mathscr{A}^{\text{samp}}) \ge \tau$ if and only if $\sigma_{\min}(A) \ge \tau\, \sigma_{\max}(A)$. The conjunction of the two conditions in eq. (4) is therefore equivalent to $\sigma_{\min}(A) \geq \tau\, \sigma_{\max}(A)$, i.e. (a). ☐

3.5. **Reading the candidate bandwidth.** The candidate definition eq. (8) is a numerical quantity attached to the sheaf $\mathscr{A}^{\mathrm{samp}}$ via the operator A. Three observations describe its character.
(i) Range. $\mathrm{sheafBW}^{\mathrm{cand}}(\mathscr{A}^{\mathrm{samp}}) \in [0, 1]$, with the value 1 attained when A is a positive scalar multiple of an isometry on V_B (the maximally well-conditioned case; cf. Theorem 5.19) and the value 0 attained when A is rank-deficient (failure of H^0 vanishing).
(ii) Three readings, one inequality. Theorem 3.2 identifies the three vocabularies in which $\sigma_{\min}(A)/\sigma_{\max}(A) \geq \tau$ is read; the candidate definition eq. (8) translates the FEEC reading verbatim into the sheaf-cohomological reading. The TopSP reading via Riesz frame bounds is recovered through clause (b) of Theorem 3.1.
(iii) Lack of categorical intrinsicness. Theorem 3.3 uses the basis $\{v_l\}_{l=0}^{K-1}$ of V_B implicit in the matrix form of A, i.e. data of the representation $V_B \hookrightarrow \mathbb{C}^N$ rather than data intrinsic to the cellular sheaf $\mathscr{A}^{\mathrm{samp}}$. The categorically intrinsic alternative, defined in section 5 as a spectral gap ratio of the sheaf coboundary operator, refers only to $\mathscr{A}$, not to an ambient embedding or to a chosen basis of stalks.

3.6. **Robinson's open problem: precise narrowing.** Robinson [Rob14b, §6] writes, in the context of cellular sheaves whose stalks are Hilbert spaces and whose cochain complexes admit Hodge decompositions, that "there may be a way to discuss the concept of *bandwidth* for general sheaves of Hilbert spaces." Theorem 3.1 together with Theorem 3.3 narrows this to a two-clause open problem.

- (R1) *(Categorical intrinsicness.)* Define a sheaf bandwidth $\mathrm{sheafBW}(\mathscr{A})$ for cellular sheaves $\mathscr{A}$ of Hilbert spaces using only the cochain-complex data of $\mathscr{A}$ (stalks, restriction maps, coboundary), without reference to an ambient embedding $V_B \hookrightarrow \mathbb{C}^N$ or to a chosen basis of stalks.
- (R2) *(Reduction on the sampling instance.)* Whatever intrinsic definition is given in (R1), it must reduce to $\sigma_{\min}(A)/\sigma_{\max}(A)$ on the sampling sheaf instance, so that the candidate eq. (8) is recovered as a special case.

The narrowing replaces the qualitative "find an equivalent definition" in [Rob14b, §6] with a concrete two-clause specification. Section 5 answers (R1) by the spectral-gap ratio definition $\mathrm{sheafBW} := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$, and answers (R2) by Theorem 5.19, which establishes the strict identity $\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) = \sigma_{\min}(A)/\sigma_{\max}(A)$ on the abstract sampling cell complex of section 5.

3.7. **Edge of soundness.** Theorem 3.1 is stated and proved within the finite-dimensional setting of section 3.1: V is a finite vertex set in Ω_k, $V_B \subset \mathbb{C}^N$ is finite dimensional, and A acts between finite-dimensional Hilbert spaces. Four boundary conditions delimit the scope of the theorem; each is recorded as a project-internal limitation, fully resolved within this paper or instantiated in a subsequent section.
(L1) Infinite-dimensional stalks. Robinson's open problem in [Rob14b, §6] is stated for sheaves of Hilbert spaces with potentially infinite-dimensional stalks, in the setting of Hilbert complexes [BL92]. The triangle equivalence is established here under the finite-dimensional restriction $\dim V_B = K < \infty$. The categorically intrinsic version of section 5 is constructed in the same finite-dimensional restriction, in which closedness of cochain ranges is automatic (see Theorem 5.10). The infinite-dimensional extension is outside the scope of this paper.
(L2) Categorical intrinsicness of the bandwidth. Theorem 3.3 is not categorically intrinsic to the sheaf $\mathscr{A}^{\mathrm{samp}}$: the basis $\{v_l\}_{l=0}^{K-1}$ underlying the matrix form of A is not part of the cellular sheaf data. The categorically intrinsic version is given in section 5; it answers clause (R1) of section 3.6 and is shown to satisfy clause (R2) on $\mathscr{A}^{\mathrm{samp}}$ via Theorem 5.19.

(L3) Spectral gap as data, not hypothesis. The bandlimited subspace $V_B(V) = \operatorname{span}\{v_l : \mu_l \le B\}$ presupposes a separating gap in the spectrum of L_0 at B; otherwise the cutoff is not stable under perturbations of V. In the finite-dimensional finite vertex setting, the spectral gap is a property of the data (V, B) rather than a hypothesis of Theorem 3.1: the theorem is stated for fixed V, V_s, B and makes no genericity assumption on the gap. The three instances of section 6 instantiate the spectral-gap data concretely; the resulting trade-off between sampling and integration is recorded in section 6 as Empirical Proposition 8b.

(L4) Vertex geometry not made explicit. The vertex geometry of V enters Theorem 3.1 only through the dependence of $V_B(V)$ on V (and through it the matrix $A = [v_l(i_n)]$). The geometry is not an independent variable of the theorem statement. The three instances of section 6 instantiate the abstract setting in three domain-specific geometries (simplicial complexes for TopSP, finite-element meshes for FEM-EM, Brillouin-zone tetrahedra for condensed matter), and Empirical Proposition 8b records the Pareto trade-off between sampling and integration across geometries within each instance.

Theorem 3.1 establishes the bridge as a formal equivalence and narrows Robinson's open problem to clauses (R1) and (R2) of section 3.6. The remaining work—constructing the categorically intrinsic sheaf bandwidth that answers (R1)—requires the FEEC machinery, which we review in section 4 before resuming the construction in section 5.

4. Finite Element Exterior Calculus: W^0 Lift and P Projection

The technical core of Robinson's open problem—a sheaf-categorically intrinsic bandwidth—requires lifting Hilbert-complex structure, Hodge decomposition, and bounded cochain projection from the continuous setting of finite element exterior calculus (FEEC) to cellular sheaves of Hilbert spaces. We review the necessary FEEC machinery in this section. The exposition is comparative: four schools—mimetic discretization [BH06], finite element systems [Chr10], FEEC Hilbert complexes [AFW10], and dagger Galerkin [LS20]— have developed the same family of constructions in disjoint vocabularies over three decades. *This section is a comparative review; we do not introduce new theory.* Its purpose is disambiguation and a side-by-side mapping that makes the sheaf-side interface of section 5 statable.

4.1. Bochev–Hyman reduction and reconstruction.

The mimetic-discretization framework of Bochev–Hyman [BH06] organizes the connection between continuous differential forms and discrete cochains around a pair of maps and a commuting diagram.

Definition 4.1 (Reduction and Whitney reconstruction)**.** Let $\mathcal{T}$ be a cellular complex and $\Lambda^k(\Omega)$ the space of differential k-forms on a domain $\Omega \supset |\mathcal{T}|$. The *reduction* (or de Rham map)

$$R : \Lambda^k(\Omega) \longrightarrow C^k(\mathcal{T}), \qquad R(\omega)(\sigma) := \int_\sigma \omega \quad \text{for } k\text{-cells } \sigma \in \mathcal{T},$$

is the integration cochain map [BH06, §1], and the *Whitney reconstruction*

$$I_W : C^k(\mathcal{T}) \longrightarrow V_W^k \subset \Lambda^k(\Omega),$$

sends a cochain to the unique Whitney k-form interpolating it on each cell of $\mathcal{T}$ [Whi57, Bos88].[2]

Proposition 4.2 (Commuting Stokes; cf. [BH06, §2])**.** *With R and I_W as above, $R \circ d = \delta \circ R$, where d is the exterior derivative on $\Lambda^k(\Omega)$ and δ is the cochain coboundary on $C^k(\mathcal{T})$. Equivalently, R is a chain map between the de Rham complex of Ω and the cochain complex of $\mathcal{T}$. The natural inner product on $C^k(\mathcal{T})$ induced by Whitney reconstruction is $\langle c, c'\rangle_{C^k} := \langle I_W c, I_W c'\rangle_{L^2(\Omega)}$.*

The pair (R, I_W) behaves asymmetrically with respect to composition:

[2]Bochev–Hyman [BH06] originally write R/I for the reduction/reconstruction pair. To avoid clash with the integration map I used in section 6 for the sampling/integration duality, we write R/I_W throughout this paper, with subscript W indicating Whitney reconstruction.

Remark 4.3 (I_W is not a projection)**.** On $V_W^k \subset \Lambda^k(\Omega)$ one has $R \circ I_W = \mathrm{id}_{C^k}$ [Whi57], but on the larger ambient $\Lambda^k(\Omega)$ the composition $I_W \circ R$ is not the identity, and I_W is not a bounded L^2-projection onto V_W^k. The same observation is recorded explicitly by Falk–Winther [FW14]: "the operator is not a projection since it is not equal to the identity on the Whitney forms." The repair is the construction of a *bounded cochain projection* π_h in section 4.3.

4.2. **Christiansen finite element systems.** The mimetic framework of section 4.1 acquires a contravariant functor form in Christiansen's finite element system (FES) framework [Chr10].

Definition 4.4 (Finite element system; [Chr10, §2, Def. 2])**.** Let $\mathcal{T}$ be a cellular complex. A *finite element system* on $\mathcal{T}$ assigns, to each $k \in \mathbb{N}$ and each cell $T \in \mathcal{T}$, a vector space $\mathcal{E}^k(T) \subseteq \widehat{X}^k(T)$ of *differential k-elements*[3] such that

(1) $d : \mathcal{E}^k(T) \to \mathcal{E}^{k+1}(T)$ is well defined; and
(2) for every cell inclusion $i : T' \subseteq T$, the pullback $i^\star : \mathcal{E}^k(T) \to \mathcal{E}^k(T')$ is a morphism of vector spaces.

Theorem 4.4 exhibits an FES as a contravariant functor from the poset category of $\mathcal{T}$ (objects: cells; morphisms: inclusions) to Vect, with $T \mapsto \mathcal{E}^k(T)$ and inclusions sent to pullbacks. Christiansen and Rapetti [CR25] state this categorical reading explicitly: "a FES is just a contravariant functor from the (poset) category determined by the mesh, to the category of vector spaces." Whitney k-forms are an instance: [Chr10, Example 2.3] identifies the case of the trimmed polynomial complex of degree 1 with the Whitney form construction of [Whi57], applied in computational electromagnetics by [Bos88].

The de Rham map of an FES recovers cohomology in the following sense.

Proposition 4.5 ([Chr10, Prop. 2.10])**.** *Under the compatibility hypotheses of* [Chr10, §2, Def. 3]*, the de Rham map from an FES to the simplicial cochain complex of $\mathcal{T}$ induces an isomorphism on cohomology.*

4.3. **AFW Hilbert complexes and bounded cochain projection.** The framework of sections 4.1 and 4.2 acquires its analytic completion in the FEEC Hilbert-complex framework of Arnold–Falk–Winther [AFW06, AFW10], which extends the algebraic structure to closed densely-defined operators between Hilbert spaces.

Definition 4.6 (Hilbert complex; [AFW10, §3.1.3], [BL92])**.** A *Hilbert complex* is a sequence $(W^k, d^k)_{k\geq 0}$ in which each W^k is a Hilbert space and each $d^k : W^k \to W^{k+1}$ is a closed, densely defined linear operator satisfying $\mathrm{range}(d^k) \subseteq \mathrm{dom}(d^{k+1})$ and $d^{k+1} \circ d^k = 0$. The complex is *closed* when, for each k, the range of d^k is closed in W^{k+1}.

Closedness yields the standard Hodge decomposition. Writing $(d^k)^*$ for the Hilbert adjoint of d^k, set $\Delta_k := d^{k-1}(d^{k-1})^* + (d^k)^* d^k$.

Theorem 4.7 (Hodge decomposition; [AFW10, §3.2],[BL92])**.** *For a closed Hilbert complex (W^k, d^k),*

$$W^k \;=\; \overline{\mathrm{range}(d^{k-1})} \;\oplus\; \ker(\Delta_k) \;\oplus\; \overline{\mathrm{range}((d^k)^*)},$$

the three summands being mutually orthogonal. The harmonic space $\mathcal{H}^k := \ker(\Delta_k)$ is canonically isomorphic to the cohomology H^k of the complex.

The discrete-side structure that connects W^k to a finite-dimensional subcomplex W_h^k is the *bounded cochain projection.*

Definition 4.8 (Bounded cochain projection; [AFW10, §3.1.1],[FW14])**.** A family of operators $\pi_h^k : W^k \to W_h^k$ is a *bounded cochain projection* when, for each k, it satisfies

[3] We follow [Chr10, §2, Def. 2] with one notational change: we write $\mathcal{E}^k(T)$ for what [Chr10] denotes $A^k(T)$, since A is reserved throughout this paper for the sampling operator $A : V_B \to \mathbb{C}^M$ of eq. (2).

(1) $\pi_h^k \circ \iota_h = \mathrm{id}_{W_h^k}$, where $\iota_h : W_h^k \hookrightarrow W^k$ is the canonical inclusion (true projection);
(2) $d^k \circ \pi_h^k = \pi_h^{k+1} \circ d^k$ on $\mathrm{dom}(d^k)$ (commutation with the coboundary);
(3) $\|\pi_h^k\|$ is uniformly bounded with respect to the mesh parameter h.

Theorem 4.9 ([AFW10, Thm. 3.6])**.** *If (W_h^k, d^k) is a subcomplex of a closed Hilbert complex (W^k, d^k) admitting a bounded cochain projection, then the cohomology of the subcomplex is canonically isomorphic to that of the ambient complex, and the harmonic spaces have equal dimension.*

Falk–Winther [FW14] construct a *local* bounded cochain projection meeting all three conditions of Theorem 4.8 on simplicial subcomplexes; the construction repairs the obstruction in Theorem 4.3 by replacing I_W with a smoothed extension that retains the commutation with d while becoming a true projection. Christiansen [Chr10, §4] gives a related family of quasi-interpolators.

4.4. **Lahtinen–Stenvall dagger mono form of Galerkin.** A complementary categorical reading is provided by Lahtinen and Stenvall [LS20], who characterize Galerkin discretization as a morphism in the dagger category Hilb of Hilbert spaces with bounded linear maps.

A *dagger category* [HV19] is a category equipped with an involutive contravariant identity-on-objects functor $\dagger$; in Hilb, $f^\dagger$ is the Hilbert adjoint of $f : H_1 \to H_2$. A *dagger mono* is a morphism $\iota : H_h \to H$ with $\iota^\dagger \circ \iota = \mathrm{id}_{H_h}$, equivalently an isometric embedding.

Proposition 4.10 (Galerkin as dagger mono; [LS20])**.** *Discretization of a finite element method, viewed as a morphism in* Hilb*, is a dagger mono $\iota_h : V_h \hookrightarrow V$ from a finite-dimensional discrete domain into the continuous Hilbert space. The Galerkin projection $V \to V_h$ is the dagger $\iota_h^\dagger$ of this embedding.*

Theorem 4.10 aligns the FEEC framework with the dagger-categorical formulation: ι_h is a dagger mono in Hilb in the same sense as the inclusion of the Whitney form subspace $V_W^k \hookrightarrow \Lambda^k(\Omega)$, and the Galerkin projection is its Hilbert adjoint. The bounded cochain projection π_h of Theorem 4.8 is not, in general, a dagger projection (i.e., it need not equal $\iota_h^\dagger$), since the local construction of [FW14] sacrifices self-adjointness in exchange for locality and uniform boundedness.

4.5. **Four-school comparison table.** The four schools of sections 4.1 to 4.4 develop the same family of constructions in disjoint vocabularies. Table 1 records the side-by-side mapping.

Construction	Bochev–Hyman [BH06]	Christiansen [Chr10]	AFW [AFW10]
Continuous space	$\Lambda^k(\Omega)$	$\widehat{X}^k(T)$	W^k
Discrete space	$C^k(\mathcal{T})$	$\mathcal{E}^k(T)$	W_h^k
Coboundary	δ on C^k	d on $\mathcal{E}^k$	d^k on W^k
Lift to ambient	I_W (Whitney)	FES inclusions [Chr10, Ex. 2.3]	$\iota_h : W_h^k \hookrightarrow W^k$
Reduction to discrete	R (de Rham map)	pullbacks $i^\star$	π_h^k (bounded cochain projection)
Commutation	$R \circ d = \delta \circ R$	functoriality of FES	$d^k \circ \pi_h^k = \pi_h^{k+1} \circ d^k$
Cohomology agreement	(implicit via Whitney 1957)	[Chr10, Prop. 2.10]	[AFW10, Thm. 3.6]

TABLE 1. The mimetic, FES, and AFW frameworks for the same family of constructions, mapped column by column. The Lahtinen–Stenvall framework [LS20] reads the inclusion row as a dagger mono in Hilb.

Remark 4.11 (Common scope)**.** Each of the four schools establishes its constructions for a single Hilbert complex (continuous side) related to a single subcomplex (discrete side). None of the four schools attaches the constructions to a sheaf over a cell complex with non-trivial restriction maps between stalks. The transition to the sheaf side is taken in section 5.

4.6. **Robinson's open problem: technical interface.** Robinson [Rob14b, §6] writes that "If F and S are sheaves of Hilbert spaces then their cochain complexes are Hilbert complexes, and they have a Hodge decomposition... There may be a way to discuss the concept of *bandwidth* for general sheaves of Hilbert spaces." Theorems 3.1 and 3.3 narrowed this to the two-clause problem (R1)–(R2) of section 3.6. Section 5 answers it in three steps, each of which uses one ingredient of the FEEC machinery reviewed above.

Step 1: *Hilbert complex structure on cellular sheaves of Hilbert spaces.* Each stalk $\mathscr{A}(\sigma)$ becomes a Hilbert space, restriction maps become bounded operators, and the cochain spaces $C^k(\mathscr{A})$ assemble into a closed Hilbert complex in the sense of Theorem 4.6. In the finite-dimensional setting of section 5, closedness of cochain ranges is automatic (Theorem 5.10). This step lifts Theorem 4.6 from a single chain to a sheaf-valued chain.

Step 2: *Hodge decomposition stalk-wise with restriction commutation.* Theorem 4.7 applied to the sheaf-cochain Hilbert complex of Step 1 yields a decomposition of $C^k(\mathscr{A})$ into exact, harmonic, and co-exact summands; the restriction maps of $\mathscr{A}$ commute with the decomposition by functoriality. The sheaf Laplacian of [HG19b] is the finite-dimensional specialization.

Step 3: *Categorically intrinsic sheaf bandwidth.* Theorem 4.8's view of well-conditioning as the boundedness of π_h is replaced on the sheaf side by a spectral-gap ratio of the sheaf coboundary δ^0, namely $\mathrm{sheafBW} := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$, defined intrinsically on the cellular sheaf data of $\mathscr{A}$. This step answers clauses (R1) and (R2) of section 3.6; it is constructed in section 5.

Steps 1 and 2 are direct transcriptions of [AFW10] Theorems 4.6 and 4.7 into the sheaf setting and contribute no new theory. Step 3 is the categorically intrinsic bandwidth definition; it is the locus of the new content of this paper, completed in section 5.

Theorems 4.6 to 4.8, together with the FES contravariant functor form Theorems 4.4 and 4.5 and the dagger formulation Theorem 4.10, supply the algebraic machinery: closed Hilbert complexes, Hodge decompositions, and bounded cochain projections, with the categorical structures that make their sheaf-side transcription unambiguous. What is missing on the sheaf side—a categorically intrinsic bandwidth answering clauses (R1) and (R2) of section 3.6—is constructed in section 5.

5. Sheaf Bandwidth: A Categorically Intrinsic Definition

Building on the FEEC machinery of section 4, we now construct the missing piece: a categorically intrinsic candidate for sheaf bandwidth, prove its invariance under unitary equivalence, and establish a strict identity reducing it to a singular value ratio on a canonical abstract sampling cell complex. The construction answers clauses (R1) and (R2) of section 3.6: the definition refers only to the cellular-sheaf data of $\mathscr{A}$ (Theorems 5.1 and 5.5), and on the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ it agrees with $\sigma_{\min}(A)/\sigma_{\max}(A)$ strictly (Theorem 5.19). Throughout this section all stalks are finite dimensional; the infinite-dimensional extension is outside the scope of this paper, in line with limitation (L1) of section 3.7.

5.1. Key design choice: $\mathrm{sheafBW} := \sqrt{\gamma_+/\Gamma_+}$.

The candidate of Theorem 3.3 expressed sheaf bandwidth via the basis-dependent quantity $\sigma_{\min}(A)/\sigma_{\max}(A)$. We introduce a categorically intrinsic alternative, formulated as a normalized spectral-gap ratio of the sheaf coboundary.

Definition 5.1 (Sheaf bandwidth)**.** Let $\mathscr{A}$ be a finite-dimensional cellular sheaf of Hilbert spaces on a finite regular cell complex X (Theorem 5.7 below), and let δ^0 be the 0-th coboundary of its cellular cochain complex (Theorem 5.9). The *sheaf bandwidth* of $\mathscr{A}$ is

$$\mathrm{sheafBW}(\mathscr{A}) \;:=\; \sqrt{\frac{\gamma_+(\delta^0)}{\Gamma_+(\delta^0)}} \;\in\; [0,1], \tag{9}$$

where $\gamma_+(\delta^0)$ and $\Gamma_+(\delta^0)$ are the infimum of the positive part of the spectrum and the supremum of the spectrum of $(\delta^0)^*\delta^0$, defined formally in Theorem 5.3. By convention, $\mathrm{sheafBW}(\mathscr{A}) := 0$

when $\delta^0 \equiv 0$ (the degenerate case in which $\mathscr{A}$ has no coboundary action; the zero map yields zero conditioning).

The choice of eq. (9) is governed by three considerations.
(a) Square root. γ_+ and Γ_+ are squared singular values of δ^0. On the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ of Theorem 5.14 below, $\gamma_+(\delta^0) = \sigma_{\min}(A)^2$ and $\Gamma_+(\delta^0) = \sigma_{\max}(A)^2$ (Theorem 5.19); the square root produces a length-scale ratio rather than an area ratio, aligning the bandwidth with the dimensionless conditioning $\sigma_{\min}/\sigma_{\max}$ that frame theory takes as the standard frame-bound ratio.
(b) Ratio. The ratio is scale invariant: global rescaling $\mathscr{A}' = c \cdot \mathscr{A}$ by a single scalar $c \in \mathbb{R} \setminus \{0\}$ applied to every stalk leaves γ_+/Γ_+ unchanged, so $\mathrm{sheafBW}(\mathscr{A}') = \mathrm{sheafBW}(\mathscr{A})$. This is the categorical requirement that bandwidth distinguish cellular sheaves up to an overall scalar.
(c) Numerator and denominator. The numerator γ_+ measures the smallest non-trivial action of δ^0, increasing with well-conditioned sheaves; the denominator Γ_+ normalizes the result to $[0, 1]$. The maximally well-conditioned case ($\mathrm{sheafBW} = 1$) corresponds to δ^0 proportional to a partial isometry on $(\ker \delta^0)^\perp$. On the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ under the hypothesis $H^0(\mathscr{A}^{\mathrm{samp}}) = 0$ of Theorem 5.19, $(\ker \delta^0)^\perp = V_B$ and this reduces to A being a positive scalar multiple of an isometry on V_B (cf. Theorem 5.19).

Table 2 positions Theorem 5.1 alongside the spectral-gap measure of Hansen and Ghrist [HG19b] and the Galerkin singular value spectrum of Lahtinen and Stenvall [LS20].

Quantity	Form	Range / dimension
$\mathrm{sheafBW}(\mathscr{A})$ (this paper)	$\sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$	dimensionless, $[0, 1]$
Sheaf spectral gap [HG19b]	$\inf \mathrm{spec}(L^k) \cap (0, \infty)$	absolute, $(0, \infty)$
Galerkin singular spectrum [LS20]	$\mathrm{spec}(\delta^*\delta)^{1/2}$	set-valued, singular values
Frame bound ratio (frame theory)	A/B on V_B	dimensionless, $[0, 1]$

TABLE 2. Comparison of Theorem 5.1 with related quantities. The sheaf bandwidth coincides with the frame-bound ratio of frame theory and is the dimensionless normalization of the spectral gap of [HG19b].

The remainder of this section makes Theorem 5.1 formally precise, proves its unitary invariance, and reduces it to $\sigma_{\min}(A)/\sigma_{\max}(A)$ on a canonical sampling sheaf.

5.2. **Sheaf Laplacian and spectral gap.** We make precise the spectral quantities entering Theorem 5.1.

Definition 5.2 (Sheaf Laplacian; cf. [HG19b, §3], [AFW10])**.** For a finite-dimensional cellular sheaf $\mathscr{A}$ on a finite regular cell complex X, with cochain complex $(C^\bullet(\mathscr{A}), \delta^\bullet)$ of Theorems 5.8 and 5.9 below, the *sheaf Laplacian* of degree k is

$$L^k(\mathscr{A}) \;:=\; \delta^{k-1}(\delta^{k-1})^* + (\delta^k)^*\delta^k \;:\; C^k(\mathscr{A}) \longrightarrow C^k(\mathscr{A}),$$

with $\delta^{-1} := 0$. By Theorem 5.10, $L^k(\mathscr{A})$ is a finite-dimensional self-adjoint positive operator.

Definition 5.3 (Spectral gap)**.** For $\mathscr{A}$ as in Theorem 5.2 and any k for which $\delta^k \neq 0$,

$$\gamma_+(\delta^k) \;:=\; \inf\{\, \lambda \in \mathrm{spec}((\delta^k)^*\delta^k) : \lambda > 0 \,\},$$
$$\Gamma_+(\delta^k) \;:=\; \sup \mathrm{spec}((\delta^k)^*\delta^k) \;=\; \|\delta^k\|_{\mathrm{op}}^2.$$

When $\delta^k = 0$ the positive part of $\mathrm{spec}((\delta^k)^*\delta^k)$ is empty and $\gamma_+(\delta^k)$ is left undefined; the degenerate case is handled directly by the convention in Theorem 5.1.

Lemma 5.4 (Well-definedness)**.** *The quantities $\gamma_+(\delta^k)$ and $\Gamma_+(\delta^k)$ depend only on the categorical data of $\mathscr{A}$ (stalks and restriction maps); they are independent of any choice of basis on the stalks.*

Proof. The operator $(\delta^k)^*\delta^k$ is self-adjoint and positive on the finite-dimensional Hilbert space $C^k(\mathscr{A})$. Since $C^k(\mathscr{A})$ is finite-dimensional, the spectrum coincides with the eigenvalue set of the matrix representation in any orthonormal basis [RS80, §VI.1, §VII.1], and this eigenvalue set is invariant under unitary change of basis. Since the inner product on $C^k(\mathscr{A})$ is determined stalk-wise by the Hilbert structure on each $\mathscr{A}(\sigma)$ (and not by a chosen basis), the spectrum—and hence γ_+, Γ_+—is determined by the sheaf data alone. □

5.3. **Unitary invariance.** The categorical-intrinsicness clause of Theorem 5.4 extends to invariance under unitary equivalence in the category of cellular sheaves of Hilbert spaces.

Proposition 5.5 (Unitary invariance of sheafBW)**.** [4] *Let $\mathscr{A}, \mathscr{A}'$ be finite-dimensional cellular sheaves of Hilbert spaces on X, and suppose there exists a* unitary *sheaf morphism $\Phi : \mathscr{A} \to \mathscr{A}'$, i.e., a family of stalk-wise unitaries $\Phi_\sigma : \mathscr{A}(\sigma) \to \mathscr{A}'(\sigma)$ satisfying the cellular-sheaf compatibility $\Phi_\tau \circ \mathscr{A}(\sigma \le \tau) = \mathscr{A}'(\sigma \le \tau) \circ \Phi_\sigma$ for every $\sigma \le \tau$. Then*

$$\mathrm{sheafBW}(\mathscr{A}) \;=\; \mathrm{sheafBW}(\mathscr{A}'). \tag{10}$$

Proof. The family $\{\Phi_\sigma\}$ assembles into a unitary operator $U : C^\bullet(\mathscr{A}) \to C^\bullet(\mathscr{A}')$ (a direct sum of stalk-wise unitaries is unitary on the Hilbert direct sum). The compatibility relation $\Phi_\tau \circ \mathscr{A}(\sigma \le \tau) = \mathscr{A}'(\sigma \le \tau) \circ \Phi_\sigma$ implies $U \circ \delta^k_{\mathscr{A}} = \delta^k_{\mathscr{A}'} \circ U$ on each C^k. Taking adjoints, $U \circ (\delta^k_{\mathscr{A}})^*(\delta^k_{\mathscr{A}}) \circ U^* = (\delta^k_{\mathscr{A}'})^*(\delta^k_{\mathscr{A}'})$ on $C^k(\mathscr{A}')$. Unitary conjugation preserves the spectrum [RS80, §VI.1], so $\gamma_+(\delta^k_{\mathscr{A}}) = \gamma_+(\delta^k_{\mathscr{A}'})$ and $\Gamma_+(\delta^k_{\mathscr{A}}) = \Gamma_+(\delta^k_{\mathscr{A}'})$ for every k, whence eq. (10). □

Remark 5.6 (Extent of invariance)**.** Theorem 5.5 establishes invariance under the *unitary subcategory* of cellular sheaves of Hilbert spaces (Theorem 5.22). Whether the invariance extends to a non-trivial dagger structure on the full category is recorded as an independent open problem (Theorem 5.25).

5.4. **Closed Hilbert complex structure.** We now record the structural ingredients required by Theorems 5.2 and 5.3: the cellular cochain complex of a finite-dimensional cellular sheaf is a closed Hilbert complex in the sense of Theorem 4.6.

Definition 5.7 (Finite-dimensional cellular sheaf of Hilbert spaces; cf. [HG19b, §2, Def. 4], [Rob14b])**.** Let X be a finite regular cell complex with face poset $(P_X, \le)$. A *finite-dimensional cellular sheaf of Hilbert spaces $\mathscr{A}$* on X assigns

(a) to each cell $\sigma \in P_X$, a finite-dimensional Hilbert space $\mathscr{A}(\sigma)$;
(b) to each $\sigma \le \tau$ in P_X, a linear operator $\mathscr{A}(\sigma \le \tau) : \mathscr{A}(\sigma) \to \mathscr{A}(\tau)$ (automatically bounded, by finite-dimensionality [RS80, §III.5]);

satisfying the functoriality $\mathscr{A}(\sigma \le \rho) = \mathscr{A}(\tau \le \rho) \circ \mathscr{A}(\sigma \le \tau)$ for $\sigma \le \tau \le \rho$.

Definition 5.8 (Cochain spaces and coboundary)**.** For $\mathscr{A}$ as in Theorem 5.7, the *cochain spaces* are

$$C^k(\mathscr{A}) \;:=\; \bigoplus_{\substack{\sigma \in P_X \\ \dim \sigma = k}} \mathscr{A}(\sigma),$$

the Hilbert direct sums of the stalks of dimension k.

Definition 5.9 (Coboundary)**.** For $\mathscr{A}$ as in Theorem 5.7, the *coboundary* $\delta^k : C^k(\mathscr{A}) \to C^{k+1}(\mathscr{A})$ is defined by

$$(\delta^k c)(\tau) \;:=\; \sum_{\substack{\sigma : \dim \sigma = k \\ \sigma < \tau}} [\sigma : \tau] \cdot \mathscr{A}(\sigma \le \tau)(c(\sigma))\,,$$

[4] Formalized in Lean 4 as `sheafBW_eq` in `lean/MRI/Cellular/Unitary.lean`. The Lean statement proves the stalk-wise unitary-morphism core on the project formalization's `Complex1D` cellular-sheaf structure for δ^0; the broader categorical packaging of finite regular cell complexes is kept informal in section 5.7. Full correspondence: `lean/CROSSREF.md`.

for $c \in C^k(\mathscr{A})$ and τ a $(k+1)$-cell, where $[\sigma : \tau] \in \{\pm 1\}$ are the incidence coefficients of the oriented cell complex (see [Hat02, §2.1]).

Theorem 5.10 (Closed Hilbert complex; cf. [AFW10, §3.1.3])**.** [5] *For $\mathscr{A}$ a finite-dimensional cellular sheaf of Hilbert spaces on a finite regular cell complex, the cochain complex $(C^\bullet(\mathscr{A}), \delta^\bullet)$ is a closed Hilbert complex in the sense of Theorem 4.6: each $C^k(\mathscr{A})$ is a finite-dimensional Hilbert space; each δ^k is a bounded linear operator; $\delta^{k+1} \circ \delta^k = 0$; and each* $\operatorname{range}(\delta^k)$ *is closed in $C^{k+1}(\mathscr{A})$.*

Proof. Each $C^k(\mathscr{A})$ is a finite Hilbert direct sum of finite-dimensional Hilbert spaces and is therefore a finite-dimensional Hilbert space. Each δ^k is a linear operator between finite-dimensional Hilbert spaces and is automatically bounded [RS80, §III.5]. The condition $\delta^{k+1} \circ \delta^k = 0$ follows from the standard incidence identity $\sum_{\tau : \sigma < \tau < \rho} [\sigma : \tau][\tau : \rho] = 0$ [Hat02, §2.1]. Range closedness is immediate: any subspace of a finite-dimensional Hilbert space is closed. □

Lemma 5.11 (Spectral form of L^0)**.** [6] *For $\mathscr{A}$ as in Theorem 5.7, $L^0(\mathscr{A}) = (\delta^0)^* \delta^0$ is a self-adjoint positive operator on $C^0(\mathscr{A})$. Its spectrum is finite,* $\operatorname{spec}(L^0(\mathscr{A})) \subset [0, \|\delta^0\|^2_{\mathrm{op}}]$*, and* $\ker L^0(\mathscr{A}) = \ker \delta^0 = H^0(\mathscr{A})$.

Proof. Self-adjointness and positivity follow from the form T^*T for any operator T. Finiteness of the spectrum follows from finite-dimensionality. The kernel identity is the standard fact that $\ker(T^*T) = \ker T$ on a Hilbert space. □

5.5. **Abstract sampling cell complex and sampling sheaf.** The reduction of Theorem 5.1 to $\sigma_{\min}(A)/\sigma_{\max}(A)$ on a sampling instance requires a cellular complex on which the sampling operator A of eq. (2) arises as the coboundary δ^0. We construct this abstract complex and the associated sheaf, and verify that the construction lands in the cellular-sheaf framework of [HG19b, Rob14b].

Definition 5.12 (Abstract sampling cell complex)**.** Let $V_s = \{i_1, \ldots, i_M\}$. The *abstract sampling cell complex* X^A is the finite regular cell complex with

- two 0-cells, σ_B and σ_∞;
- M one-cells $\tau_1, \ldots, \tau_M$, with each τ_i having boundary $\partial \tau_i = \{\sigma_B, \sigma_\infty\}$;
- incidence coefficients $[\sigma_B : \tau_i] = +1$, $[\sigma_\infty : \tau_i] = -1$ ([Hat02, §2.1], with each τ_i oriented from σ_B to σ_∞).

The cell σ_B is the *principal* 0*-cell*; the cell σ_∞ is an *auxiliary basepoint* with no physical content—its presence ensures that each τ_i has two distinct endpoints, the regularity condition required by Theorem 5.13 below.

Remark 5.13 (Regularity verification)**.** X^A satisfies the regular cell complex conditions of [HG19b, §2, Def. 1], requiring for each cell of dimension n_α a homeomorphism of a closed ball in $\mathbb{R}^{n_\alpha}$ onto the closed cell, mapping the open ball homeomorphically onto the cell. Each 0-cell is a single point, which is the closed unit ball in $\mathbb{R}^0$ and satisfies the condition trivially. For each 1-cell τ_i, the closed unit ball in $\mathbb{R}^1$ is the interval $[0, 1]$, whose boundary $S^0 = \{0, 1\}$ must map to two *distinct* 0-cells of X^A; the assignment $0 \mapsto \sigma_B$, $1 \mapsto \sigma_\infty$ realizes the required homeomorphism. The double 0-cell construction is what verifies *Cond. 4* of [HG19b, §2, Def. 1] (boundaries of 1-cells must consist of

[5] Formalized in Lean 4 at the `Complex1D` / δ^0 layer: `coboundaryHilbert_range_isClosed` in `lean/MRI/Cellular/HilbertComplex.lean` proves condition (d), range closedness. Conditions (a)–(b) are finite-dimensional auto-derivations and condition (c) is vacuous in the one-dimensional complex used by the formalization; the all-k finite-regular-cell-complex statement is not packaged as one Lean theorem. Full correspondence: `lean/CROSSREF.md`.

[6] Formalized in Lean 4: the substantive operator claims correspond to three separate Lean theorems in `lean/MRI/Cellular/HilbertComplex.lean`: `sheafLaplacian0_isSymmetric`, `sheafLaplacian0_isPositive`, and `sheafLaplacian0_ker_eq`. The finite-spectrum and norm-bound consequences are standard finite-dimensional/self-adjoint/positive-operator facts supplied by mathlib rather than separate project theorems. Full correspondence: `lean/CROSSREF.md`.

two distinct 0-cells); Cond. (1)–(3) hold automatically by the cell-complex setup of Theorem 5.12. Functoriality is vacuous on a 1-dimensional cell complex (there are no chains of three distinct cells with strict inclusions). Hence X^A is a regular cell complex in the sense of [HG19b, §2, Def. 1, Cond. 4].

Definition 5.14 (Sampling sheaf)**.** The *sampling sheaf* $\mathscr{A}^{\mathrm{samp}}$ on X^A is the cellular sheaf defined by

- $\mathscr{A}^{\mathrm{samp}}(\sigma_B) := V_B$ (with the inner product inherited from $\mathbb{C}^N$);
- $\mathscr{A}^{\mathrm{samp}}(\sigma_\infty) := 0$ (the zero Hilbert space);
- $\mathscr{A}^{\mathrm{samp}}(\tau_i) := \mathbb{C}$ for each $i \in V_s$;
- the restriction maps are $\mathscr{A}^{\mathrm{samp}}(\sigma_B \leq \tau_i) := \mathrm{eval}_i$, where $\mathrm{eval}_i : V_B \to \mathbb{C}$, $c \mapsto c_i$, is evaluation at the i-th index, and $\mathscr{A}^{\mathrm{samp}}(\sigma_\infty \leq \tau_i) := 0$, the unique linear map from the zero space to $\mathbb{C}$.

Remark 5.15 (Cellular-sheaf framework)**.** $\mathscr{A}^{\mathrm{samp}}$ satisfies the cellular sheaf axioms of [HG19b, §2, Def. 4]: each stalk is a finite-dimensional Hilbert space (the zero space being the zero object of $\mathsf{Hilb}^{\mathrm{fin}}$, the category of finite-dimensional Hilbert spaces); each restriction map is a bounded linear operator between finite-dimensional Hilbert spaces; functoriality is vacuous since X^A has no chains of three distinct cells with strict inclusions.

Lemma 5.16 (Canonical isometry $V_B \oplus 0 \cong V_B$)**.** *The Hilbert direct sum $V_B \oplus \{0\} = \{(c, 0) : c \in V_B\}$ is canonically isometric to V_B via the projection $\pi_B(c, 0) := c$, with inverse $\iota_B(c) := (c, 0)$. Both maps are isometries: $\|(c, 0)\|^2_{V_B \oplus 0} = \|c\|^2_{V_B}$* [RS80, §II.1].

Lemma 5.17 (Cochain spaces of $\mathscr{A}^{\mathrm{samp}}$)**.** *For $\mathscr{A}^{\mathrm{samp}}$ as in Theorem 5.14,*

$$C^0(\mathscr{A}^{\mathrm{samp}}) \;=\; V_B \oplus \{0\} \;\cong\; V_B \qquad \text{and} \qquad C^1(\mathscr{A}^{\mathrm{samp}}) \;=\; \bigoplus_{i \in V_s} \mathbb{C} \;\cong\; \mathbb{C}^M,$$

the first isomorphism being the canonical isometry of Theorem 5.16.

Proof. $C^0(\mathscr{A}^{\mathrm{samp}})$ is the direct sum of the stalks on 0-cells: $\mathscr{A}^{\mathrm{samp}}(\sigma_B) \oplus \mathscr{A}^{\mathrm{samp}}(\sigma_\infty) = V_B \oplus 0$, and Theorem 5.16 provides the canonical isometry to V_B. $C^1(\mathscr{A}^{\mathrm{samp}})$ is the direct sum of the stalks on 1-cells, each a copy of $\mathbb{C}$, indexed by V_s. □

Proposition 5.18 (Coboundary equals sampling operator)**.** [7] *Under the canonical isometry of Theorems 5.16 and 5.17, the coboundary $\delta^0 : C^0(\mathscr{A}^{\mathrm{samp}}) \to C^1(\mathscr{A}^{\mathrm{samp}})$ of $\mathscr{A}^{\mathrm{samp}}$ agrees with the sampling operator $A : V_B \to \mathbb{C}^M$ of eq. (2):*

$$\delta^0 \;=\; A \qquad \text{as linear maps, after the canonical identification of Theorems 5.16 and 5.17.} \tag{11}$$

The identification is canonical (uniquely determined by the sheaf data), so no choice is involved in the comparison.

Proof. By Theorems 5.9 and 5.14, for $(c, 0) \in V_B \oplus 0$,

$$\begin{aligned}
\big(\delta^0(c, 0)\big)(\tau_i) &= [\sigma_B : \tau_i] \cdot \mathscr{A}^{\mathrm{samp}}(\sigma_B \leq \tau_i)(c) + [\sigma_\infty : \tau_i] \cdot \mathscr{A}^{\mathrm{samp}}(\sigma_\infty \leq \tau_i)(0) \\
&= (+1) \cdot \mathrm{eval}_i(c) + (-1) \cdot 0 \\
&= c_i
\end{aligned}$$

for each $i \in V_s$. Stacking over $i \in V_s$, $\delta^0(c, 0) = (c_{i_1}, \ldots, c_{i_M}) = Ac$. Composing with the canonical isometry of Theorem 5.16 (which is itself an isometry, hence does not change operator norms or spectra), $\delta^0 = A$ holds as linear maps after the canonical identification. □

[7] Formalized in Lean 4 by two statements in `lean/MRI/Cellular/Sampling.lean`: `coboundary_eq_sampling_op` proves the operator-layer equality after the canonical map $E \to V_B \oplus 0$, and `coboundaryHilbert_eq_samplingOpEuclidean` proves the Hilbert-layer/PiLp version used for singular values. Full correspondence: `lean/CROSSREF.md`.

5.6. **Strict equality on the sampling sheaf.** The reduction of Theorem 5.1 on the sampling sheaf is now a direct consequence of Theorem 5.18.

Theorem 5.19 (Strict equality)**.** [8] *Let V, V_s, B be as in section 3.1 (and hence $V_B = V_B(V) \subset \mathbb{C}^N$ and $A : V_B \to \mathbb{C}^M$ as in eq. (2)). Assume A has full column rank, equivalently $\sigma_{\min}(A) > 0$, equivalently $H^0(\mathscr{A}^{\mathrm{samp}}) = 0$ (the three formulations are mutually equivalent by Theorem 5.20). For the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ on the abstract sampling cell complex X^A of Theorems 5.12 and 5.14,*

$$\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) \;=\; \frac{\sigma_{\min}(A)}{\sigma_{\max}(A)} \;\in\; (0,1]. \tag{12}$$

The upper boundary $\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) = 1$ *is attained iff $\sigma_{\min}(A) = \sigma_{\max}(A)$, i.e., iff A is a positive scalar multiple of an isometry on V_B (cf. design clause (c) of Theorem 5.1).*

Proof. By Theorem 5.18, $\delta^0 = A$ as an operator $V_B \to \mathbb{C}^M$ (under the canonical isometry $C^0(\mathscr{A}^{\mathrm{samp}}) \cong V_B$). The spectrum of $(\delta^0)^*\delta^0$ therefore equals the spectrum of A^*A, which is the set of squared singular values of A: $\mathrm{spec}(A^*A) = \{\sigma_l(A)^2\}_{l=0}^{K-1}$. Hence

$$\begin{aligned}\gamma_+(\delta^0) &= \inf\{\, \lambda \in \mathrm{spec}(A^*A) : \lambda > 0 \,\} = \sigma_{\min}(A)^2,\\ \Gamma_+(\delta^0) &= \sup \mathrm{spec}(A^*A) = \sigma_{\max}(A)^2.\end{aligned}$$

Substituting into eq. (9),

$$\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) \;=\; \sqrt{\frac{\gamma_+(\delta^0)}{\Gamma_+(\delta^0)}} \;=\; \sqrt{\frac{\sigma_{\min}(A)^2}{\sigma_{\max}(A)^2}} \;=\; \frac{\sigma_{\min}(A)}{\sigma_{\max}(A)},$$

each equality being a strict equality of real numbers (no asymptotic or almost-equal qualification); the operator-level identification $\delta^0 = A$ used at the start of the proof is canonical (Theorem 5.18). ☐

Corollary 5.20 (H^0 vanishing on the sampling sheaf)**.** [9] $H^0(\mathscr{A}^{\mathrm{samp}}) = \ker A$. *In particular,* $H^0(\mathscr{A}^{\mathrm{samp}}) = 0$ *if and only if* $\sigma_{\min}(A) > 0$.

Proof. $H^0(\mathscr{A}^{\mathrm{samp}}) = \ker \delta^0$ by the standard cellular-sheaf identification of 0-cohomology with global sections; by Theorem 5.18, $\ker \delta^0 = \ker A$, and $\ker A = 0$ iff $\sigma_{\min}(A) > 0$. ☐

Remark 5.21 (Rank-deficient locus)**.** When $\sigma_{\min}(A) = 0$ (equivalently $H^0(\mathscr{A}^{\mathrm{samp}}) \neq 0$), the hypothesis of Theorem 5.19 fails and the two sides of eq. (12) disagree: the candidate $\mathrm{sheafBW}^{\mathrm{cand}}(\mathscr{A}^{\mathrm{samp}})$ of Theorem 3.3 is set to 0 by convention, while the intrinsic invariant $\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) = \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$ measures the spectral-gap ratio of δ^0 restricted to $(\ker \delta^0)^\perp$ and remains strictly positive whenever $\delta^0 \neq 0$ on $(\ker \delta^0)^\perp$. The two quantities encode different information on the rank-deficient locus: the candidate records by convention the failure of full sampling, the intrinsic invariant records the conditioning of δ^0 on its non-trivial range. Theorem 5.19 is the statement that on the well-conditioned locus $\sigma_{\min}(A) > 0$ the two quantities coincide strictly.

---

[8] Formalized in Lean 4: the paper's full-column-rank form corresponds to `strict_equality_full_rank` in `lean/MRI/Cellular/SamplingSpectral.lean` (the equality $\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}) = \sigma_{\min}(A)/\sigma_{\max}(A)$ with domain-last $\sigma_{\min}$ and the interval $(0,1]$); the upper-boundary iff corresponds to `sheafBW_Samps_eq_one_iff_singularValues_domain_eq`. The helper `strict_equality` proves the more general positive-spectrum version indexed by `finrank (range A) - 1`. The further geometric interpretation "iff A is a positive scalar multiple of an isometry on V_B" is a corollary of $\sigma_{\min}(A) = \sigma_{\max}(A)$ derived in the proof body below and is not separately formalized as a Lean theorem. Full correspondence: `lean/CROSSREF.md`.

[9] Formalized in Lean 4: the two statements of this corollary correspond to (i) `globalSections_comap_eq_ker_samplingOp` in `lean/MRI/Cellular/TriangleEquivalence.lean` (the equality $H^0(\mathscr{A}^{\mathrm{samp}}) = \ker A$ expressed as a `Submodule.comap` via the canonical isometry); and (ii) `globalSections_eq_bot_iff_singularValues_last_pos` in the same file (the iff $H^0(\mathscr{A}^{\mathrm{samp}}) = 0 \iff \sigma_{\min}(A) > 0$). The intermediate `H0_vanishing_iff` in `lean/MRI/Cellular/SamplingSpectral.lean` bridges to the `Injective` form. Full correspondence: `lean/CROSSREF.md`.

Theorem 5.19 answers clause (R2) of section 3.6 in the well-conditioned regime: the categorically intrinsic sheaf bandwidth of Theorem 5.1 reduces to the conditioning ratio $\sigma_{\min}(A)/\sigma_{\max}(A)$ on the sampling sheaf with strict equality. Theorem 5.21 delineates the rank-deficient boundary of the strict equality.

5.7. **Unitary subcategory and dagger structure.** Theorem 5.5 required the existence of a unitary sheaf morphism between $\mathscr{A}$ and $\mathscr{A}'$. We make the categorical setting explicit and identify its dagger structure.

Definition 5.22 (Unitary subcategory)**.** [10] Let X be a finite regular cell complex. The category $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)$ has finite-dimensional cellular sheaves of Hilbert spaces on X as objects, and stalk-wise families of linear operators $\Phi = \{\Phi_\sigma\}_{\sigma\in P_X}$ satisfying the cellular-sheaf compatibility as morphisms. Its *unitary subcategory*, $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb},\mathrm{u}}(X)$, is the subcategory whose morphisms are those Φ with each Φ_σ a unitary operator.

Proposition 5.23 (Unitary subcategory is a groupoid)**.** [11] *The category* $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb},\mathrm{u}}(X)$ *is a groupoid. On this subcategory the dagger functor* $\Phi^\dagger := \{(\Phi_\sigma)^*\}_{\sigma\in P_X}$ *coincides with the inverse functor:* $\Phi^\dagger = \Phi^{-1}$.

Proof. Let $\Phi : \mathscr{A} \to \mathscr{A}'$ be unitary, so each $\Phi_\sigma : \mathscr{A}(\sigma) \to \mathscr{A}'(\sigma)$ is a unitary operator on a finite-dimensional Hilbert space; in particular $\Phi_\sigma^{-1} = \Phi_\sigma^*$ [RS80, §VI.1]. To show Φ is invertible in $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb},\mathrm{u}}(X)$, we verify that the family $\{\Phi_\sigma^{-1}\}_{\sigma\in P_X}$ defines a sheaf morphism. The sheaf morphism axiom for Φ reads $\Phi_\tau \circ \mathscr{A}(\sigma \le \tau) = \mathscr{A}'(\sigma \le \tau) \circ \Phi_\sigma$ for every $\sigma \le \tau$. Applying Φ_τ^{-1} on the left and Φ_σ^{-1} on the right yields $\mathscr{A}(\sigma \le \tau) \circ \Phi_\sigma^{-1} = \Phi_\tau^{-1} \circ \mathscr{A}'(\sigma \le \tau)$, the sheaf morphism axiom for $\{\Phi_\sigma^{-1}\}$ in the opposite direction. Each Φ_σ^{-1} is unitary (inverse of a unitary is unitary), so $\Phi^{-1} \in \mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb},\mathrm{u}}(X)$. The identity $\Phi_\sigma^\dagger = \Phi_\sigma^* = \Phi_\sigma^{-1}$ extends stalk-wise to $\Phi^\dagger = \Phi^{-1}$. The category is therefore a groupoid [ML71, §I.5]. $\square$

Remark 5.24 (Dagger structure on the unitary subcategory is trivial)**.** On a groupoid the dagger functor of Theorem 5.23 *coincides* with the inverse functor; the dagger axioms of [HV19, §2.3] are then satisfied by the groupoid structure alone. This corresponds to the *dagger-core* of a †-category in the standard nomenclature: it is the maximal sub-groupoid on which dagger and inverse agree. The substantive content of dagger category theory—the distinction of dagger from inverse on non-isomorphism morphisms (isometries, partial isometries, dagger-monics, dagger-kernels)—is absent on the unitary subcategory by construction.

5.8. **Open Problem: non-trivial dagger on the full category.** The trivial dagger structure on the unitary subcategory of Theorem 5.24 leaves open the question of whether a non-trivial dagger structure exists on the full category $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)$.

Open Problem 5.25 (Non-trivial dagger on $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)$)**.** Does there exist a contravariant involutive functor

$$\dagger \;:\; \mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X) \longrightarrow \mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)^{\mathrm{op}}$$

satisfying the standard dagger-functor axioms of [HV19, §2.3, 2.18–2.19], namely

(D1) $\mathscr{A}^\dagger = \mathscr{A}$ for every object;
(D2) $(\mathrm{id}_{\mathscr{A}})^\dagger = \mathrm{id}_{\mathscr{A}}$;
(D3) $(\Phi \circ \Psi)^\dagger = \Psi^\dagger \circ \Phi^\dagger$;

---

[10] Not formalized in Lean 4: the categorical packaging of the unitary subcategory awaits a fuller cellular-sheaf category in mathlib. The core stalk-wise unitary structure used in the invariance proposition above is captured by `UnitarySheafMorphism` in `lean/MRI/Cellular/Unitary.lean`; see `lean/CROSSREF.md`.

[11] Not formalized in Lean 4; the groupoid/dagger structure on the unitary subcategory is left informal pending a fuller categorical formalization in mathlib. See `lean/CROSSREF.md`.

(D4) $(\Phi^\dagger)^\dagger = \Phi$;

that restricts on the unitary subcategory $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb,u}}(X)$ to the stalk-wise Hilbert adjoint of Theorem 5.23, and is moreover *non-trivial* on non-unitary morphisms in the sense of distinguishing isometries from dagger-monics from arbitrary morphisms?

The obstruction to a stalk-wise definition is recorded for context. On a non-unitary morphism Φ, the stalk-wise Hilbert adjoints $\{(\Phi_\sigma)^*\}_{\sigma\in P_X}$ in general fail the cellular sheaf morphism axiom: starting from $\Phi_\tau \circ \mathscr{A}(\sigma\le\tau) = \mathscr{A}'(\sigma\le\tau)\circ\Phi_\sigma$ and taking adjoints yields $\mathscr{A}(\sigma\le\tau)^*\circ\Phi_\tau^* = \Phi_\sigma^*\circ\mathscr{A}'(\sigma\le\tau)^*$, which is the sheaf morphism axiom for the *adjoint restriction maps* $\mathscr{A}(\sigma\le\tau)^*, \mathscr{A}'(\sigma\le\tau)^*$ running in the opposite direction in the face poset. Without unitarity of restriction maps—in general absent in cellular sheaves of Hilbert spaces—the stalk-wise dagger fails to be a sheaf morphism. A non-trivial dagger functor on the full category must therefore use a non-stalk-wise extension, with no precedent in the cellular-sheaf literature: [LS20] treats dagger structure on the category of finite element discretizations (distinct from cellular sheaves), and [HG19b] introduces the unitary subcategory but does not extend a dagger beyond it.

Theorem 5.25 is independent of the main results of this paper. Theorems 5.5 and 5.19 hold under unitary equivalence in $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb,u}}(X)$, which is sufficient for the intrinsicness clause (R1) of section 3.6 in the form proven here. A non-trivial dagger on the full category would strengthen Theorem 5.5 to dagger invariance, but is not required for either the strict equality of Theorem 5.19 or for the three-instance instantiation of section 6.

Theorems 5.1, 5.5 and 5.19 answer clauses (R1) and (R2) of section 3.6: the sheaf bandwidth is defined intrinsically from the cellular-sheaf data, is invariant under unitary equivalence, and reduces to $\sigma_{\min}(A)/\sigma_{\max}(A)$ on the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ with strict equality. To exhibit the framework's reach beyond the canonical sampling instance, we now apply it to three previously disjoint domains—signal processing on simplicial complexes, finite element electromagnetics, and Brillouin zone tetrahedron quadrature—unified by a common categorical template.

6. Three Instances: TopSP, Finite Element EM, Brillouin Zone

We instantiate the framework of section 5 in three previously disjoint domains. Each instance specializes the abstract domain $V_?$ to a domain-specific Hilbert subspace and the abstract operator A to a domain-specific evaluation or quadrature operator, while the categorical structure—the abstract sampling cell complex X^A (Theorem 5.12), the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ (Theorem 5.14), the strict equality Theorem 5.19, and the unitary invariance Theorem 5.5—is shared verbatim across the three instances. The categorical structure is unified verbatim, but the *physical interpretations* differ instance by instance: in particular, the sampling map S and integration map I introduced in section 6.6 are formal scalar maps on the geometry data V, and the role they play (evaluation versus quadrature, real versus synthetic data, independence versus coupling of S and I) is instance-specific. The *verbatim* clause refers to the categorical structure only, not to the physical correspondence; the explicit divergences are recorded in table 3 and in the per-instance discussion of section 6.4.

6.1. **Common setup.** Throughout this section we fix a finite-dimensional Hilbert subspace $V_? \subset \mathbb{C}^N$ of dimension K, a sampling location set $V_s \subset \{1,\dots,N\}$ of size M, and the corresponding sampling operator

$$A : V_? \to \mathbb{C}^M, \qquad c \mapsto \left(c_{i_n}\right)_{n=1}^{M},$$

where $\{i_n\}_{n=1}^M$ enumerates V_s. The abstract sampling cell complex X^A is defined by Theorem 5.12 (one principal 0-cell σ_B, an auxiliary 0-cell σ_∞, and M 1-cells $\{\tau_i\}_{i\in V_s}$). The sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ is defined by Theorem 5.14 with stalks $\mathscr{A}^{\mathrm{samp}}(\sigma_B) = V_?$, $\mathscr{A}^{\mathrm{samp}}(\sigma_\infty) = 0$, $\mathscr{A}^{\mathrm{samp}}(\tau_i) = \mathbb{C}$, and restriction maps eval_i at σ_B and zero at σ_∞.

By Theorem 5.19, under the hypothesis $\sigma_{\min}(A) > 0$ (equivalently $H^0(\mathscr{A}^{\mathrm{samp}}_{V_s,V_?}) = 0$),

$$\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}_{V_s,V_?}) \;=\; \frac{\sigma_{\min}(A)}{\sigma_{\max}(A)}, \tag{13}$$

and by Theorem 5.5 this quantity is invariant under unitary equivalence of cellular sheaves over X^A. Each of the three instances below verifies the full-column-rank hypothesis under its standard configuration: Instance 1 by direct numerical check on 35 MRI configurations (section 6.2), Instance 2 by the H^1-Rayleigh basis of V_S^{FEM} being linearly independent at the chosen vertex sample (section 6.3), and Instance 3 by non-degeneracy of the linear tetrahedron weight matrix w^{lin} on the occupied subspace (section 6.4).

The three instances differ only in the choice of $(V_?, A)$. We display them in a uniform format and summarize the comparison in table 3.

6.2. **Instance 1: Topological signal processing.**

Domain V_B^{TopSP}. Let $X(V)$ be a cellular complex with vertex set $V = \{k_1, \dots, k_N\} \subset \Omega_k$, and let $L_0 = D - W$ be its graph Laplacian with eigendecomposition $(\mu_l, v_l)_{l=0}^{N-1}$, $\mu_0 \le \cdots \le \mu_{N-1}$. Following [BS20, §4.4], the Hodge bandlimited subspace at threshold B is

$$V_B^{\mathrm{TopSP}} \;=\; \mathrm{span}\{\, v_l : \mu_l \le B \,\} \;\subset\; \mathbb{C}^N, \qquad K \;=\; \dim V_B^{\mathrm{TopSP}}.$$

Operator A^{TopSP}. With sampling location set V_s of size M, the sampling operator is

$$A^{\mathrm{TopSP}} \;:\; V_B^{\mathrm{TopSP}} \to \mathbb{C}^M, \qquad A^{\mathrm{TopSP}}_{n,l} \;=\; v_l(i_n), \qquad n = 1, \dots, M, \;\; l = 0, \dots, K-1.$$

Sheaf bandwidth. By eq. (13),

$$\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}_{V_s,V_B^{\mathrm{TopSP}}}) \;=\; \frac{\sigma_{\min}(A^{\mathrm{TopSP}})}{\sigma_{\max}(A^{\mathrm{TopSP}})}.$$

Empirical evidence on real MRI data. A simulation suite developed in the present project applied $\sigma_{\min}(A^{\mathrm{TopSP}})/\sigma_{\max}(A^{\mathrm{TopSP}})$ to 35 configurations covering five trajectory families (Cartesian, random, sparkling, spiral, radial) and seven V_s sizes (M ranging from 30 to 400, with base scale $N = 200$). All 35 configurations attained full column rank, with a clear ordering: Cartesian sampling at sheafBW ≈ 0.0091 versus radial sampling at sheafBW $\approx 2.07 \times 10^{-5}$, separating the two regimes by more than two orders of magnitude. The ordering matches the reconstruction quality reported in MRI sampling practice [LDP07, LWC+19]. The numerical study is reported in section 7.

6.3. **Instance 2: Finite element electromagnetics.**

Setup. Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain with a simplicial mesh $K(\tau_h)$ (each top-dimensional simplex a tetrahedron). Following [Whi57, Bos88], the Whitney 0-form space is

$$W^0(K) \;=\; \mathrm{span}\{\, \lambda_i : i = 1, \dots, N \,\} \;\cong\; \mathbb{C}^N \qquad (\lambda_i \text{ the hat function at vertex } x_i).$$

Domain V_S^{FEM}. Let $\{w_l\}_{l=0}^{K-1}$ be the lowest K eigenmodes of the H^1-Rayleigh quotient on $W^0(K)$, normalized so that $\langle w_l, w_m \rangle_{L^2} = \delta_{lm}$:

$$\langle \nabla w_l, \nabla \varphi \rangle_{L^2} \;=\; \lambda_l \, \langle w_l, \varphi \rangle_{L^2} \quad \forall \varphi \in W^0(K), \qquad \lambda_0 \le \cdots \le \lambda_{K-1}.$$

Define

$$V_S^{\mathrm{FEM}} \;:=\; \mathrm{span}\{\, w_l : l = 0, \dots, K-1 \,\} \;\subset\; \mathbb{C}^N.$$

Then V_S^{FEM} is a K-dimensional Hilbert subspace of $\mathbb{C}^N$.

Operator A^{FEM}. Choose M vertices $\{x_{i_n}\}_{n=1}^{M}$ as the sampling locations. The sampling operator is

$$A^{\mathrm{FEM}} : V_S^{\mathrm{FEM}} \to \mathbb{C}^M, \qquad (A^{\mathrm{FEM}} c)_n \;=\; \sum_{l=0}^{K-1} c_l\, w_l(x_{i_n}),$$

i.e. $A^{\mathrm{FEM}}_{n,l} = w_l(i_n)$, the same matrix form as A^{TopSP} with the basis $\{v_l\}$ replaced by the H^1-Rayleigh modes $\{w_l\}$. The pair $(V_S^{\mathrm{FEM}}, A^{\mathrm{FEM}})$ satisfies the input requirements of Theorem 5.19: (i) V_S^{FEM} is a finite-dimensional Hilbert subspace by construction; (ii) A^{FEM} is bounded and linear, automatic in finite dimensions; (iii) $\sigma_{\min}(A^{\mathrm{FEM}}) > 0$ holds under the standard non-degeneracy condition that the H^1-Rayleigh modes $\{w_l\}_{l=0}^{K-1}$ are linearly independent on the chosen vertex sample $\{x_{i_n}\}_{n=1}^{M}$, a mild assumption on the mesh and sample configuration.

Project-internal scope: relation to AFW bounded cochain projection. [AFW10] construct on the continuous Hilbert complex $(W^k(\Omega), d^k)$ a bounded cochain projection π_h satisfying $\pi_h \circ \iota = \mathrm{id}_{V_h}$ and $d \circ \pi_h = \pi_h \circ d$ ([AFW10, §3.1.1]). The operator A^{FEM} above and the AFW projector π_h are distinct objects:

	A^{FEM} (this paper)	π_h ([AFW10])
Domain	$V_S^{\mathrm{FEM}} \subset \mathbb{C}^N$	$H\Lambda^k(\Omega)$
Codomain	$\mathbb{C}^M$ (vertex evaluations)	finite-dim V_h
Role	sampling operator	projection / interpolation

The hypothesis of Theorem 5.19 requires a finite-dimensional Hilbert subspace and a bounded linear operator; it does not require a bounded cochain projection. Instance 2 is established within the finite-dimensional restriction of section 5.4; the continuous-side extension to $H^1(\Omega)$ via π_h is outside the scope of the present paper.

Empirical evidence. Instance 2 is established as a *scope-bounded categorical claim*: the construction lies within the finite-dimensional restriction discussed above, and the symbolic identification $A^{\mathrm{FEM}} \cong \pi_h$ holds verbatim under that restriction. No independent empirical channel (numerical study or established practice) is provided for Instance 2 in this paper; the evidence channels listed in table 4 record this explicitly. The numerical verification in section 7 covers the algebraic identity eq. (13) on Instance 1 and does not probe Instance 2 separately. The continuous-side extension to $H^1(\Omega)$ via the bounded cochain projection π_h is outside the present paper's scope and is recorded as an engineering outlook in section 8.

Sheaf bandwidth. By eq. (13),

$$\mathrm{sheafBW}(\mathscr{A}^{\mathrm{samp}}_{V_s, V_S^{\mathrm{FEM}}}) \;=\; \frac{\sigma_{\min}(A^{\mathrm{FEM}})}{\sigma_{\max}(A^{\mathrm{FEM}})}.$$

6.4. **Instance 3: Brillouin zone tetrahedron quadrature.**

Setup. Let $\Omega_{\mathrm{BZ}} \subset \mathbb{R}^3$ be the first Brillouin zone of a crystalline solid, equipped with a tetrahedron tessellation by reciprocal-space points $\{\mathbf{k}_i\}_{i=1}^{N}$. For a fixed band index, let ϵ_i be the band energy at $\mathbf{k}_i$ and ϵ_F the Fermi level.

Domain $V_{\mathrm{occ}}^{\mathrm{BZ}}$. In the linear-segment regime of the Blöchl tetrahedron method [BJA94, §II.B], the band energy $\epsilon(\mathbf{k})$ is linearly interpolated within each tetrahedron and the Heaviside step $\theta(\epsilon_F - \epsilon)$ is integrated exactly over this linear segment, yielding the linear tetrahedron weights $w^{\mathrm{lin}}_{i_n,j}$ of [LT72, JA71, BJA94]. We define

$$V_{\mathrm{occ}}^{\mathrm{BZ}} \;:=\; \mathrm{span}\{\, \chi_i : i \in \mathrm{occ}(\epsilon_F)\,\} \;\subset\; \mathbb{C}^N$$

as the span of Whitney 0-form vertex hat functions χ_i at $\mathbf{k}_i$ over occupied k-points $\mathrm{occ}(\epsilon_F) = \{i : \epsilon_i \le \epsilon_F\}$; $K = |\mathrm{occ}(\epsilon_F)|$.

Scope of $V_{\text{occ}}^{\text{BZ}}$. The construction of $V_{\text{occ}}^{\text{BZ}}$ as a Whitney-0-form span is a project-internal categorical reformulation of the linear-tetrahedron framework; it is *not* a standard "occupied subspace" of band theory (the standard object is the span of Bloch eigenfunctions of occupied bands, residing in a distinct vector space). The reformulation is motivated by the fact that the linear tetrahedron weights $w_{i_n,j}^{\text{lin}}$ of [BJA94, LT72, JA71] are defined on this hat-function basis, making $V_{\text{occ}}^{\text{BZ}}$ the natural domain for the operator A^{BZ} of the next paragraph.

Operator A^{BZ}. With M test points (a sub-tetrahedral integration grid) at vertex locations,

$$A^{\text{BZ}} : V_{\text{occ}}^{\text{BZ}} \to \mathbb{C}^M, \qquad A_{n,j}^{\text{BZ}} \;=\; w_{i_n j}^{\text{lin}},$$

where $w_{i_n j}^{\text{lin}}$ is the linear tetrahedron weight of [LT72, JA71].

Physical role of A^{BZ} versus A^{TopSP} and A^{FEM}. Across the three instances, the abstract operator $A : V_? \to \mathbb{C}^M$ of section 6.1 plays structurally different physical roles. In sections 6.2 and 6.3 the matrix A is a *row-sampling* operator at vertex locations: $A_{n,l}^{\text{TopSP}} = v_l(i_n)$ and $A_{n,l}^{\text{FEM}} = w_l(i_n)$ are bare evaluations of basis functions. In Instance 3 the matrix $A_{n,j}^{\text{BZ}} = w_{i_n j}^{\text{lin}}$ is the linear-tetrahedron *integration weight* matrix: the n-th row records the weights $w_{i_n j}^{\text{lin}}$ that the sub-tetrahedral integration grid attaches to the occupied vertex basis at test point i_n. The two roles—row-sampling and quadrature weighting—are physically distinct.

The categorical framework of section 5, however, requires only that A be a bounded linear map $V_? \to \mathbb{C}^M$; it does not distinguish sampling-style from quadrature-style operators. Consequently:

- the strict equality eq. (13) applies to A^{BZ} verbatim, with sheafBW measuring the conditioning of the integration weight matrix on $V_{\text{occ}}^{\text{BZ}}$;
- the sampling map S and the integration map I defined in Theorems 6.1 and 6.2 are formal scalar maps on the geometry parameter V and do not depend on the physical role of A within an instance. In Instance 3, S^{BZ} and I^{BZ} both involve the integration weight matrix A^{BZ} and are therefore not physically independent within the BZ context, although they remain formally distinct as scalar maps of V. The directional content of Theorem 6.3 is established across instances on the V-parameter side, not within Instance 3 in isolation.

Project-internal scope: Blöchl correction. [BJA94] introduces a quadratic correction term that improves the tetrahedron quadrature beyond the linear segment. This correction is a Richardson-extrapolation-style refinement that does not lie in the Whitney 0-form framework ([BJA94, §II.C]). Instance 3 is established within the linear-segment regime; extending sheaf bandwidth analysis to the Blöchl-corrected quadrature is outside the scope of the present paper.

Sheaf bandwidth. By eq. (13),

$$\text{sheafBW}(\mathscr{A}_{V_s, V_{\text{occ}}^{\text{BZ}}}^{\text{samp}}) \;=\; \frac{\sigma_{\min}(A^{\text{BZ}})}{\sigma_{\max}(A^{\text{BZ}})}.$$

Empirical evidence: five decades of practice. Production density-functional codes have, since 1971, used non-Cartesian tessellations (Monkhorst–Pack [MP76]; tetrahedron methods [LT72, JA71, BJA94]) for Brillouin zone integration, while Cartesian quadrature is documented to perform worst on smooth band-structure integrands. This empirical hierarchy is consistent with the Pareto trade-off articulated below (Theorem 6.3).

6.5. **Three-instance comparison.**

6.6. **Sampling and integration maps.** The unified description suggests two scalar maps on the same data.

	Instance 1: TopSP	**Instance 2: FEM–EM**	**Instance 3: BZ**
$V_?$	V_B^{TopSP} (Hodge band)	V_S^{FEM} (H^1-Rayleigh)	$V_{\text{occ}}^{\text{BZ}}$ (Whitney 0-form, occupied)
$A_{n,l}$	$v_l(i_n)$	$w_l(i_n)$	$w_{i_n j}^{\text{lin}}$
Physical role of A	row-sampling	row-sampling	integration weights
sheafBW	$\sigma_{\min}/\sigma_{\max}$	$\sigma_{\min}/\sigma_{\max}$	$\sigma_{\min}/\sigma_{\max}$
Empirical evidence	35/35 MRI configs	finite-dim restriction	50-year practice
Reference	[BS20]	[AFW10, Whi57]	[BJA94, LT72]

TABLE 3. Three instances of the framework of section 5. The categorical structure $X^A + \mathscr{A}^{\text{samp}} + \textit{Theorem } 5.19 + \textit{Theorem } 5.5$ is shared verbatim. The instances differ only in the domain-specific choice of $(V_?, A)$; the spectral ratio $\sigma_{\min}(A)/\sigma_{\max}(A)$ is the unified invariant.

Definition 6.1 (Sampling map)**.** [12] For a fixed $V_?$, the *sampling map* S assigns to a sampling cell complex X_V^A (parameterized by the geometry V) the spectral ratio

$$S(X_V^A) \;:=\; \text{sheafBW}(\mathscr{A}_V^{\text{samp}}) \;=\; \frac{\sigma_{\min}(A_V)}{\sigma_{\max}(A_V)}.$$

By Theorem 5.5, S is invariant under unitary equivalence of cellular sheaves over X^A.

Definition 6.2 (Integration map)**.** [13] For a fixed $V_?$, the *integration map* I assigns to X_V^A the quadrature error norm

$$I(X_V^A) \;:=\; \Big\| \mathcal{I}^* \;-\; \sum_{i\in V_s} w_i \;\text{eval}_i^* \Big\|_{V_? \to \mathbb{C}},$$

where $\mathcal{I}^* : V_? \to \mathbb{C}$ is an exact-integration dual functional and $\{w_i\}_{i\in V_s}$ are quadrature weights, both chosen instance-specifically (e.g., the linear-tetrahedron weight w^{lin} of section 6.4 for Instance 3).

The choice of $\mathcal{I}^*$ and $\{w_i\}$ is fixed per instance and not unique across instances; the across-instance directional content of Theorem 6.3 does not depend on a specific choice (cf. the disclaim in section 6.4 on the formal nature of S, I).

6.7. **Empirical Proposition: S–I Pareto trade-off.**

Empirical Proposition 6.3 (Sampling and integration trace a Pareto frontier under V-optimization)**.** [14] Fix the type of $V_?$ (one of V_B^{TopSP}, V_S^{FEM}, $V_{\text{occ}}^{\text{BZ}}$) and the strategy for choosing V_s. As the geometry V varies over admissible cellular complexes, no V simultaneously maximizes $V \mapsto S(X_V^A)$ and minimizes $V \mapsto I(X_V^A)$; the pair (S, I) traces a Pareto frontier in V.

Heuristic skeleton. The strict equality eq. (13) gives the S-side directly. For the I-side, the spectral framework of [PSC+15] on Euclidean point sets suggests, for sufficiently smooth integrands, a directional decomposition: the integration error scales with the high-frequency tail of the integrand expanded in the $V_?$ eigenbasis, weighted by the Fourier spectrum of the sample-set indicator. This is a heuristic of directional value only, not a quantitative bound on cellular-sheaf integration. The directional content of Theorem 6.3 is the following.

[12] Not formalized in Lean 4. The sampling map S is a physical correspondence (B-spline FE, quadrature, Bloch–Floquet ψ_θ, etc.) from geometry V to spectral data, not a mathematical theorem statement. See `lean/CROSSREF.md` §C.

[13] Not formalized in Lean 4. The integration map I is a physical correspondence (quadrature error norm) from geometry V to scalar error, not a mathematical theorem statement. See `lean/CROSSREF.md` §C.

[14] Not formalized in Lean 4. This is an *empirical* proposition supported by two quantitative evidence channels plus one heuristic skeleton (see body); it is not promoted to theorem status per the double-track verification principle. See `lean/CROSSREF.md` §C.

- S is large iff the columns of A_V are close to orthonormal on V_s, which holds when the geometry V_s is locally coherent with the $V_?$-modes $\{v_l\}$. Regular structures (Cartesian grids) achieve this local coherence.
- I is small iff the spectrum of V_s flattens at high frequencies, which requires the geometry V_s to break regular structure. Blue-noise tessellations [HB19] and Monkhorst–Pack grids [MP76] exemplify this.

The two requirements pull V in opposite directions: sampling favors regular structure; integration favors broken regularity. This is the heuristic content of Theorem 6.3.
Evidence and heuristic skeleton. The directional content of Theorem 6.3 is supported by two empirical channels and a heuristic skeleton, recorded in table 4.

Type	**Item**	**Source**	**Strength**
Empirical	b3pre numerical study (35 configs, multi-N robustness)	this paper, section 7	strong
Empirical	Brillouin zone practice (five decades)	[BJA94, LT72, JA71]	strong
Heuristic	Euclidean spectral framework (Fourier-spectrum form)	[PSC$^+$15]	directional only

Table 4. Two empirical channels and one heuristic skeleton supporting Theorem 6.3. The b3pre numerical study and the multi-N robustness scan are stratifications of a single project-internal study, counted as one empirical channel. The Brillouin zone practice channel is independent published evidence on V-geometry choice for quadrature, not on the joint sampling-vs-integration trade-off; it supports the directional content via the integration side. The Pilleboue spectral framework is a heuristic skeleton on Euclidean Monte Carlo variance, not direct evidence on cellular-sheaf integration on simplicial or tetrahedral complexes.

Status. Theorem 6.3 is a directional statement supported by two empirical channels and a heuristic skeleton (table 4); it is not a constructive Pareto-optimality theorem. Constructive proofs typically require compactness arguments and convex-optimization duality applied to the joint (S, I) map. The constructive proof is treated as an independent open problem (Theorem 6.4 below). Should that open problem be resolved, Theorem 6.3 upgrades to a theorem.

6.8. **Open Problem: constructive Pareto proof.**

Open Problem 6.4 (Constructive Pareto proof for Theorem 6.3)**.** Establish, by compactness arguments and convex-optimization duality on the joint map $V \mapsto (S(X_V^A), I(X_V^A))$, the formal statement that no admissible geometry V simultaneously achieves $S(X_V^A) > S^* - \epsilon$ and $I(X_V^A) < I^* + \epsilon$ for sufficiently small $\epsilon > 0$, where (S^*, I^*) are the Pareto extrema. The expected mode of attack:
(1) compactness of the V-parameter space (under suitable boundary conditions);
(2) convex relaxation of the joint map, with the dual optimization encoding the trade-off;
(3) closing the duality gap via spectral arguments paralleling [PSC$^+$15].

Theorem 6.4 is independent of the main theorems of this paper: Theorem 6.3 is not invoked in the proof of Theorem 5.19, and the categorical bridge stands without it.

The unified framework reduces all three instances to a spectral ratio on A. We verify this strict equality numerically across the three instances in section 7.

7. Numerical Verification

Theorem 5.19 reduces sheafBW$(\mathscr{A}^{\mathrm{samp}})$ to a spectral ratio on the sampling operator A. We verify this strict equality numerically across the three instances of section 6. The verification confirms the linear-algebraic identity and the code implementation; it does not constitute independent physical evidence for each instance, as we make explicit in section 7.3.

7.1. **Experimental setup.** A self-contained Python script (`A_route_ch3_numerical.py`, 147 lines, depending only on `numpy` [HMvdW+20]) constructs a toy operator $A \in \mathbb{C}^{M\times K}$ for each of the three instances and computes two quantities for comparison:

(i) the spectral-gap form $\sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$, where $(\delta^0)^*\delta^0 = A^*A$ on the relevant $V_?$ subspace;
(ii) the singular-value form $\sigma_{\min}(A)/\sigma_{\max}(A)$.

For each instance, A is generated by drawing a random orthonormal basis $U \in \mathbb{C}^{N\times K}$ via QR factorization of a complex Gaussian matrix and then sub-sampling M rows of U to obtain A, mirroring the row-sampled basis representation $A_{n,l} = v_l(i_n)$ used uniformly across the three instances of section 6. The dimensions are

Instance	N	K	M	Subspace label
1: TopSP	100	10	30	V_B^{TopSP}
2: FEM–EM	50	6	8	V_S^{FEM}
3: BZ	30	4	12	$V_{\mathrm{occ}}^{\mathrm{BZ}}$

The random seed is fixed at `seed` = 20260507 for reproducibility. The relative deviation between the two forms is defined as

$$\mathrm{reldiff} := \frac{\left|\sqrt{\gamma_+/\Gamma_+} - \sigma_{\min}/\sigma_{\max}\right|}{\sigma_{\min}/\sigma_{\max}}.$$

Instance	$\sqrt{\gamma_+/\Gamma_+}$	$\sigma_{\min}/\sigma_{\max}$	reldiff
1: TopSP	0.367756144311	0.367756144311	7.55×10^{-16}
2: FEM–EM	0.053301378959	0.053301378959	1.64×10^{-14}
3: BZ	0.519194438983	0.519194438983	2.14×10^{-16}

TABLE 5. Numerical verification of Theorem 5.19 on the three instances at `seed` = 20260507. The two forms (spectral-gap and singular-value) of Theorem 5.1 agree to at least twelve decimal digits in every instance. Instances 1 and 3 attain reldiff within $4 \times \epsilon_{\mathrm{mach}}$ (IEEE 754 unit roundoff $\epsilon_{\mathrm{mach}} \approx 2.22 \times 10^{-16}$); Instance 2 deviates by 1.64×10^{-14}, reflecting the expected roundoff amplification of comparing $\sqrt{\lambda_{\min}(A^*A)/\lambda_{\max}(A^*A)}$ with $\sigma_{\min}(A)/\sigma_{\max}(A)$ when A is poorly conditioned, since $\kappa(A^*A) = \kappa(A)^2$. All three deviations are at least an order of magnitude below the floating-point error budget for this comparison on matrices of the given shapes.

7.2. **Results.** The maximum relative deviation across the three instances is 1.64×10^{-14} (Instance 2), driven by the squaring of the condition number under the A^*A form rather than by any numerical failure of Theorem 5.19. The spectral-gap form and the singular-value form agree to within the floating-point error budget in every instance.

7.3. **Honest statement on the nature of this verification.** The script of section 7.1 verifies the spectral-gap form $\sqrt{\gamma_+/\Gamma_+}$ and the singular-value form $\sigma_{\min}/\sigma_{\max}$ of Theorem 5.1 agree numerically on a randomly generated A of the relevant shape. Two facts are verified:

- the linear-algebraic identity $\sqrt{\lambda_{\min}(A^*A)/\lambda_{\max}(A^*A)} = \sigma_{\min}(A)/\sigma_{\max}(A)$, which holds for any $M \times K$ matrix A (a standard consequence of the singular value decomposition);
- the code implementation (`A_route_ch3_numerical.py`) computes both forms without introducing numerical artifacts beyond IEEE 754 roundoff.

What the script does *not* do:

- It does not provide independent physical evidence that the domain $V_?$ in each instance is correctly chosen for the underlying application (signal reconstruction in TopSP, finite element solution in FEM–EM, band-structure integration in BZ). All three instances use the same toy construction $U = \mathrm{QR}(\mathrm{randn}(N, K))$, which only matches the abstract structural template "finite-dim Hilbert subspace + row-sampled basis representation."
- It does not validate the heuristic skeleton of Theorem 6.3: that proposition rests on the crossed evidence channels of table 4, not on this script.
- It does not exercise the cellular-sheaf construction of section 5 independently: by Theorem 5.18, $\delta^0 = A$ on the sampling sheaf, so $(\delta^0)^*\delta^0 = A^*A$ holds by definition rather than as a numerical fact. The deviation reported in table 5 is the IEEE 754 roundoff of computing eigenvalues of A^*A versus squared singular values of A; it does not test the cellular sheaf structure.

The script is best understood as a sanity check on the algebra of Theorem 5.1 and on the code path that computes it, not as empirical support for the framework's applicability to any specific domain.

7.4. **Physical evidence per instance.** The empirical content of each instance rests on the following sources, which are not exercised by the script of section 7.1.

Instance	Empirical content	Source
1: TopSP	35 MRI sampling configurations; full column rank in all 35; Cartesian sheafBW ≈ 0.0091 vs. radial $\approx 2.07 \times 10^{-5}$	this paper, project simulation suite (section 6.2)
2: FEM–EM	finite-dimensional restriction is a project-internal scope statement, not an empirical claim	section 6.3
3: BZ	five decades of condensed-matter practice consistently report non-Cartesian tessellations outperforming Cartesian quadrature	[BJA94, LT72, JA71, MP76]

Table 6. Sources of empirical content per instance. Instance 2 is established by the finite-dimensional restriction of section 5.4; the row labeled "finite-dimensional restriction" is a project-internal scope statement and is not a software- or hardware-derived empirical claim.

The numerical verification closes the main argument of the paper. We conclude with a discussion of the two independent open problems left by this work.

8. Discussion and Open Problems

We have constructed a categorical bridge between finite element exterior calculus, cellular sheaf signal processing, and topological signal processing, instantiated it in three previously disjoint domains, and verified its main equality numerically to machine precision. We now restate the seven contributions in the order they were established, give the precise content of the two open problems that remain, and outline the engineering verification on the MRI side that lies outside this paper.

8.1. **Summary of contributions.**

(C1) The triangle equivalence (Theorem 3.1) reads one scalar—$\sigma_{\min}(A)/\sigma_{\max}(A)$—through three schools: FEEC singular value bounds (section 3.3), TopSP Riesz frame conditions (Theorem 3.2), and Robinson sheaf cohomology (section 3.4). The proof produces a candidate sheaf bandwidth that is not yet categorically intrinsic (Theorem 3.3), and section 3.6 precisely narrows Robinson's open problem to the construction of an intrinsic version.

(C2) The categorically intrinsic candidate $\mathrm{sheafBW}(\mathscr{A}) := \sqrt{\gamma_+(\delta^0)/\Gamma_+(\delta^0)}$ (Theorem 5.1) is unitarily invariant (Theorem 5.5). The well-definedness of the spectral gap quantities relies on the closed Hilbert complex structure of Theorem 5.10, which uses the finite-dimensional restriction of Theorem 5.7.

(C3) On the abstract sampling cell complex X^A (Theorem 5.12), the sampling sheaf $\mathscr{A}^{\mathrm{samp}}$ (Theorem 5.14) reduces the intrinsic invariant to the ambient ratio $\sigma_{\min}(A)/\sigma_{\max}(A)$ *strictly* (Theorem 5.19) on the well-conditioned locus $\sigma_{\min}(A) > 0$. The double 0-cell construction $\{\sigma_B, \sigma_\infty\}$ is what places X^A into the Hansen–Ghrist regularity framework [HG19b, Def. 1, Cond. 4]; Theorem 5.21 delineates the rank-deficient boundary on which the strict equality fails.

(C4) Three previously disjoint domains (section 6) fit a common categorical template: TopSP (V_B^{TopSP}, section 6.2), FEM–EM (V_S^{FEM}, section 6.3), and Brillouin zone tetrahedron quadrature ($V_{\mathrm{occ}}^{\mathrm{BZ}}$, section 6.4). The three-instance comparison is recorded in table 3.

(C5) The empirical proposition (Theorem 6.3) formulates the S–I Pareto trade-off as a directional empirical statement supported by two empirical channels and a heuristic skeleton (table 4); it is not promoted to theorem status, and it is not invoked in the proof of Theorem 5.19—the categorical bridge is independent of it.

(C6) The numerical verification (section 7) records reldiff $\leq 1.64 \times 10^{-14}$ across the three toy instances at `seed` $= 20260507$ (with Instances 1 and 3 attaining $\leq 8 \times 10^{-16}$), with explicit scope: the script verifies the SVD-based identity and the code path, not the per-instance physical evidence (section 7.3 and table 6).

(C7) Two independent open problems remain, formulated below.

8.2. **Open Problem: non-trivial dagger on the full category.** Theorem 5.25 asks whether the unitary subcategory groupoid of Theorem 5.22 extends to a non-trivial † structure on the full category $\mathsf{Cell\text{-}Sh}^{\mathrm{fin}}_{\mathsf{Hilb}}(X)$. The substantive content of this question is the tension between two candidate lifts: dagger compactness in Hilb [HV19, §2.3] does not lift directly to cellular sheaves because the restriction maps are not in general isometries, while the dagger mono structure of [LS20, §3] does lift but produces only the unitary subgroupoid rather than a structure on the full category. Theorem 5.24 explains why the naive lift collapses; the construction of a non-trivial extension is left open. Theorem 5.25 is independent of all theorems of this paper.

8.3. **Open Problem: constructive Pareto proof.** Theorem 6.4 asks for a constructive Pareto-optimality proof upgrading the empirical proposition Theorem 6.3 to a theorem. The proposed mode of attack proceeds in three steps: compactness of the V-parameter space under suitable boundary conditions; convex relaxation of the joint map $V \mapsto (S(X_V^A), I(X_V^A))$, with the dual optimization encoding the trade-off; and closing the duality gap via spectral arguments paralleling [PSC+15]. Resolution of this problem would replace Theorem 6.3 by a theorem and the heuristic skeleton by a constructive proof. Theorem 6.4 is independent of the categorical bridge and of all theorems of this paper.

8.4. **Engineering outlook on the MRI side.** The triangle equivalence (Theorem 3.1) and the strict equality (Theorem 5.19) reduce, on the MRI sampling problem, to a singular value ratio computable directly from the acquisition matrix A. The project-internal numerical study recorded in table 4 examines 35 sampling configurations at $N = 200$; the multi-N scan ($N \in \{200, 300, 1000\}$) shows the top three configurations (Cartesian, random, sparkling) and the bottom two (spiral, radial) separated by two to three orders of magnitude in the singular value ratio across the entire range. A larger-scale engineering verification—with $N \approx 10^4$ acquisition points, real phantom data, and one representative baseline drawn from each of the four schools surveyed in section 2—is the natural next step on the MRI side. That engineering verification is outside the scope of this paper.

Acknowledgments. [Acknowledgments placeholder.]

References

[AFW06] Douglas N. Arnold, Richard S. Falk, and Ragnar Winther, *Finite element exterior calculus, homological techniques, and applications*, Acta Numerica **15** (2006), 1–155.

[AFW10] ______, *Finite element exterior calculus: From Hodge theory to numerical stability*, Bulletin of the American Mathematical Society **47** (2010), no. 2, 281–354.

[BH06] Pavel B. Bochev and James M. Hyman, *Principles of mimetic discretizations of differential operators*, Compatible Spatial Discretizations (Douglas N. Arnold, Pavel B. Bochev, Richard B. Lehoucq, Roy A. Nicolaides, and Mikhail Shashkov, eds.), The IMA Volumes in Mathematics and its Applications, vol. 142, Springer, New York, 2006, pp. 89–119.

[BJA94] Peter E. Blöchl, Ole Jepsen, and Ole K. Andersen, *Improved tetrahedron method for Brillouin-zone integrations*, Physical Review B **49** (1994), no. 23, 16223–16233.

[BL92] Jochen Brüning and Matthias Lesch, *Hilbert complexes*, Journal of Functional Analysis **108** (1992), no. 1, 88–132, Foundational treatment of Hilbert complexes; cited by Hansen-Ghrist 2019 §3.3 for the infinite-dimensional Hodge decomposition.

[Bos88] Alain Bossavit, *Whitney forms: A class of finite elements for three-dimensional computations in electromagnetism*, IEE Proceedings A (Physical Science, Measurement and Instrumentation, Management and Education) **135** (1988), no. 8, 493–500.

[BS20] Sergio Barbarossa and Stefania Sardellitti, *Topological signal processing over simplicial complexes*, IEEE Transactions on Signal Processing **68** (2020), 2992–3007.

[Chr10] Snorre H. Christiansen, *Finite element systems of differential forms*, arXiv preprint (2010), Definition 2 (FES = contravariant functor from poset of cells to vector spaces) and Proposition 2.10 used in §4. arXiv preprint posted 2010-06-24; published as Christiansen-Munthe-Kaas-Owren, Acta Numerica vol 20 (2011) — see arxiv abs page for the journal version.

[CKL25] Paula Cerejeiras, Uwe Kähler, and Dmitrii Legatiuk, *Script geometry: Categorical perspective and a connection to exterior calculus FEM*, Complex Analysis and Operator Theory **19** (2025), no. 4, Article 75, Categorical framework on tight scripts (Sommen school); does not connect to cellular sheaf signal processing.

[CR25] Snorre H. Christiansen and Francesca Rapetti, *Interpretation of a discrete de rham method as a finite element system*, arXiv preprint (2025), DDR method shown to define computable consistent discrete L2 product on conforming FES; not connected to cellular sheaf signal processing.

[FW14] Richard S. Falk and Ragnar Winther, *Local bounded cochain projections*, Mathematics of Computation **83** (2014), no. 290, 2631–2656, Establishes that Whitney interpolation is not a projection; constructs a true local bounded cochain projection.

[Hat02] Allen Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002, §2.1 for incidence coefficients and oriented cell complex conventions.

[HB19] Eric Heitz and Laurent Belcour, *Distributing monte carlo errors as a blue noise in screen space by permuting pixel seeds between frames*, Computer Graphics Forum **38** (2019), no. 4, 149–158.

[HG19a] Jakob Hansen and Robert Ghrist, *Distributed optimization with sheaf homological constraints*, 2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton), IEEE, 2019, Introduces sheaf Laplacians as drop-in replacements for graph Laplacians in distributed algorithms. (Bibkey renamed from HansenGhrist2018 — original entry's arXiv 1808.01513 was the Spectral Theory paper, not this one.), pp. 565–571.

[HG19b] ______, *Toward a spectral theory of cellular sheaves*, Journal of Applied and Computational Topology **3** (2019), no. 4, 315–358.

[HMvdW+20] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant, *Array programming with NumPy*, Nature **585** (2020), no. 7825, 357–362.

[HV19] Chris Heunen and Jamie Vicary, *Categories for quantum theory: An introduction*, Oxford Graduate Texts in Mathematics, vol. 28, Oxford University Press, Oxford, 2019, Chapter 2 (§2.3 + 2.18–2.19) for the standard dagger functor axioms.

[JA71] Ole Jepsen and Ole K. Andersen, *The electronic structure of h.c.p. ytterbium*, Solid State Communications **9** (1971), no. 20, 1763–1767.

[LDP07] Michael Lustig, David Donoho, and John M. Pauly, *Sparse MRI: The application of compressed sensing for rapid MR imaging*, Magnetic Resonance in Medicine **58** (2007), no. 6, 1182–1195.

[LS20] Valtteri Lahtinen and Antti Stenvall, *A category theoretical interpretation of discretization in Galerkin finite element method*, Mathematische Zeitschrift **296** (2020), no. 3-4, 1271–1285, Discretization characterized as a dagger mono with discrete domain in the category of Hilbert spaces.

[LT72] Gernot Lehmann and Manfred Taut, *On the numerical calculation of the density of states and related properties*, Physica Status Solidi (B) **54** (1972), no. 2, 469–477.

[LWC$^+$19] Carole Lazarus, Pierre Weiss, Nicolas Chauffert, Franck Mauconduit, Loubna El Gueddari, Christophe Destrieux, Ilyess Zemmoura, Alexandre Vignaud, and Philippe Ciuciu, *SPARKLING: Variable-density k-space filling curves for accelerated T_2^*-weighted MRI*, Magnetic Resonance in Medicine **81** (2019), no. 6, 3643–3661.

[ML71] Saunders Mac Lane, *Categories for the working mathematician*, Graduate Texts in Mathematics, vol. 5, Springer, New York, 1971, §I.5 for the standard definition of a groupoid. First edition 1971; 2nd edition 1998.

[MP76] Hendrik J. Monkhorst and James D. Pack, *Special points for Brillouin-zone integrations*, Physical Review B **13** (1976), no. 12, 5188–5192.

[PSC$^+$15] Adrien Pilleboue, Gurprit Singh, David Coeurjolly, Michael Kazhdan, and Victor Ostromoukhov, *Variance analysis for Monte Carlo integration*, ACM Transactions on Graphics **34** (2015), no. 4, 124:1–124:14, Proc. SIGGRAPH 2015.

[Rob14a] Michael Robinson, *A sheaf-theoretic perspective on sampling*, 2014, Sheaf perspective on sampling; complementary in topic to the bandwidth open problem of [Rob14b] §6.

[Rob14b] ———, *Topological signal processing*, Mathematical Engineering, Springer, Berlin, Heidelberg, 2014, §6 leaves open the question of a sheaf-categorical bandwidth definition for sheaves of Hilbert spaces.

[RS80] Michael Reed and Barry Simon, *Methods of modern mathematical physics. I: Functional analysis*, revised and enlarged ed., Academic Press, New York, 1980, §II.1 for Hilbert direct sum and zero-summand isometry $V \oplus 0 \cong V$.

[TW14] Lloyd N. Trefethen and J. A. C. Weideman, *The exponentially convergent trapezoidal rule*, SIAM Review **56** (2014), no. 3, 385–458, Use as Trefethen reference for high-order quadrature theory.

[Whi57] Hassler Whitney, *Geometric integration theory*, Princeton Mathematical Series, vol. 21, Princeton University Press, Princeton, NJ, 1957.

Independent Researcher
Email address: `[Email placeholder]`